Electrolytes :-

① **Strong Electrolytes :-** 100% ionizable

eg: St. Acid : HCl, HBr, HI, H_2SO_4, $HClO_4$, $HBrO_4$, HIO_4, HNO_3 etc

St. Base :- hydroxide of alkali metal and alkaline earth metal.

Salts :

② **Weak Electrolyte :-** Partially ionizable

eg: W. Acid : HF, HNO_2, H_3PO_2, H_3PO_3, H_3PO_4, H_2S, CH_3COOH, $HCOOH$ etc

Organic Acids generally behave as weak Acids.

eg: W. Base : NH_3 or NH_4OH, $Zn(OH)_2$, C_5H_5N, CH_3NH_2, $C_6H_5NH_2$ etc

Hydroxides of d-block generally behave as w. base

Organic bases are generally w. bases.

Strong electrolytes are more conducting than weak electrolyte in solution.

Ostwald's Dilution Law :-

A^+B^- = Binary weak Electrolyte

K = Ionization Cont

$$AB_{aq} \overset{K}{\rightleftharpoons} A^+_{aq} + B^-_{aq}$$

$t=0$ $\quad a_0$

t_{eq} $\quad a_0(1-\alpha) \quad a_0\alpha \quad a_0\alpha$

$$K = \frac{a_0\alpha^2}{(1-\alpha)}$$

For very weak electrolyte $(1-\alpha) \simeq 1$

$$K = a_0\alpha^2$$

$$\boxed{\alpha = \sqrt{\frac{K}{a_0}}}$$

$$\Rightarrow \boxed{\alpha \propto \sqrt{V}}$$

$$\Rightarrow \boxed{\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{V_1}{V_2}}}$$

$\Rightarrow V \to \infty \Rightarrow \alpha \simeq 1$

Note: If $\boxed{\frac{K}{a_0} \leq 10^{-3}}$ *

$\Rightarrow (1-\alpha) \simeq 1$

Theory of Acids and Bases :

① **Arrhenius theory :** Substance which provide H^+ to soln are Acids while which provide OH^- to soln are base.

eg: $HClO_4\ (aq) \longrightarrow H^+ + ClO_4^-$ (Acid)

$CsOH\ (aq) \longrightarrow OH^- + Cs^+$ (Base)

② **Bronsted Lowry theory :** Substance which provide H^+ to Soln are Acids.

Substance which take H^+ from soln are base.

eg: $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$

Acid Base Base Acid

CH_3COOH / CH_3COO^- Conjugate Acid/Base

H_2O / H_3O^+ Conjugate Base / Acid

① eg:

Acid	Conjugate Base
HCl	Cl^-
H_2CO_3	HCO_3^-
H_3PO_4	$H_2PO_4^-$
$H_2PO_4^-$	HPO_4^{-2}
NH_4^+	NH_3

H_3PO_4 HPO_4^{-2} ✗

② eg: $HCl > HF$ (given) $\Rightarrow Cl^- < F^-$

$HCOOH > CH_3COOH$ (given) $\Rightarrow HCOO^- < CH_3COO^-$

$C_2H_4 < C_2H_2$ (given) $\Rightarrow C_2H_3^- > C_2H^-$

③ eg:

HCl (SA)	Cl^- ✗	# Na^+Cl^- ✗ ✗	Neutral
eg: $HClO_4$ (S.A)	ClO_4^- ✗	$CH_3COO^-Na^+$ ✓ ✗	Basic
eg: $NaOH$ (S.B)	Na^+ ✗	$NH_4^+Cl^-$ ✓ ✗	Acidic
		$ZnCl_2$ ✓ ✗	Acidic

④ @ CH_3COONa

✓ ✗

$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

ⓑ NH_4Cl

✓ ✗

$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

(c) $H_3PO_3 \rightleftharpoons H^+ + H_2PO_3^-$

$H_2PO_3^- \rightleftharpoons H^+ + HPO_3^{--}$

(d) $Zn(OH)_2 \rightleftharpoons Zn^+(OH) + OH^-$

$Zn^+(OH) \rightleftharpoons Zn^{+2} + OH^-$

(e) Na_2CO_3

X ✓

$CO_3^{--} + H_2O \rightleftharpoons HCO_3^- + OH^-$

$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$

(3) Lewis Acid/Base theory :

Acid : accept e⁻ pair
Base : donate e⁻ pair

Acid :

(a) +ve charge
(b) incomplete octate
(c) Vacant d-orbital
(d) ↓

eg: CH_3^+

BH_3
↑

I_2
↑

$O=\overset{+\delta}{C}=\overset{-\delta}{O}$
↑

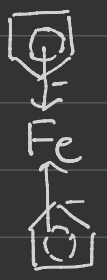
Base :

(a) -ve charge
(b) lone pair
(c)

eg: CH_3^-, OH^-, I^-

$:NH_3$, $C_5H_5N:$
↑

$(C_5H_5)_2Fe$ (Ferrocene)



Ionic Product of Water: (K_w)

$$H_2O \underset{}{\overset{K_{eq}}{\rightleftharpoons}} H^+ + OH^- \quad ; \Delta H = 57.1 \text{ KJ or } 13.7 \text{ KCal}$$

a_0

$a_0(1-\alpha) \simeq a_0 \qquad a_0\alpha \qquad a_0\alpha$

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]} \Rightarrow \boxed{[H^+][OH^-] = K_w = 10^{-14}} \text{ at RT}$$

= Ionic Product of water

$$= \frac{10^{-14}}{\left(\frac{1000}{18}\right)} = 1.8 \times 10^{-16}$$

$$K_{eq} = 1.8 \times 10^{-16} = \frac{(a_0\alpha)^2}{a_0(1-\alpha)} = a_0\alpha^2$$

$$\alpha^2 = \frac{1.8 \times 10^{-16}}{\left(\frac{1000}{18}\right)} = (1.8)^2 \times 10^{-18}$$

$$\alpha = 1.8 \times 10^{-9}$$

Note: ① $T\uparrow \Rightarrow \xrightarrow{Rxn} \Rightarrow H^+\uparrow$ & $OH^-\uparrow \Rightarrow K_w\uparrow$

② for H_2O

$$[H^+] = [OH^-]$$

$$[H^+][H^+] = 10^{-14} \Rightarrow [H^+] = 10^{-7} M = [OH^-]$$

If $[H^+] > [OH^-] \Rightarrow [H^+] > 10^{-7} M \Rightarrow$ Acidic

$[H^+] < [OH^-] \Rightarrow [H^+] < 10^{-7} M \Rightarrow$ Basic

$[H^+] = [OH^-] \Rightarrow [H^+] = 10^{-7} M \Rightarrow$ Neutral

③

Relation b/w Ka and Kb of Conjugate Acid/Base :-

HA = Weak Acid , K_a = Ionization Const
A^- = Conjugate Base , K_b = Ionization Const

$$HA \underset{}{\overset{K_a}{\rightleftharpoons}} H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]} \quad -(I)$$

$$A^- + H_2O \overset{K_b}{\rightleftharpoons} HA + OH^-$$

$$K_b = \frac{[HA][OH^-]}{[A^-]} \quad -(II)$$

Eqn (I) × (II) *

$$\boxed{K_a \times K_b = [H^+][OH^-] = K_w = 10^{-14}} \text{ at RT}$$

eg: CH_3COOH $\left(K_a = 1.8 \times 10^{-5}\right)$

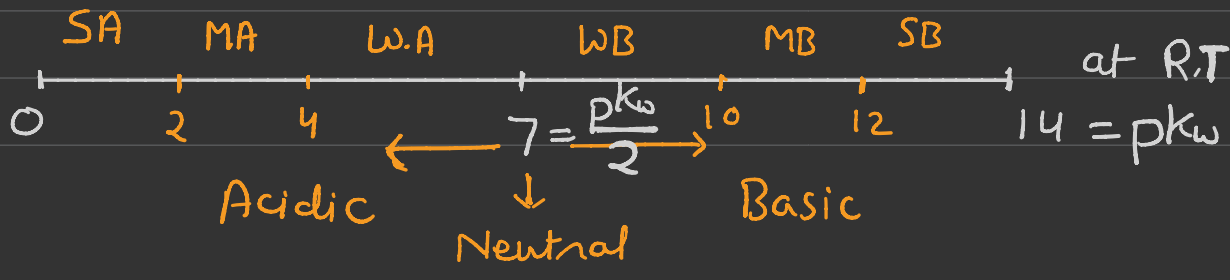
CH_3COO^- $\quad K_b = \frac{10^{-14}}{1.8 \times 10^{-5}} = \left(\frac{10}{1.8}\right) \times 10^{-10}$

pH Scale :-

① $pH = -\log[H^+]$

$pOH = -\log[OH^-]$

②



③ $pK = -\log K$

$pK_a = -\log K_a$

$pK_b = -\log K_b$

$pK_w = -\log K_w = 14$ at RT

④ $K_a \times K_b = K_w$

$\boxed{pK_a + pK_b = pK_w}$

⑤ $[H^+][OH^-] = K_w$

$\boxed{pH + pOH = pK_w = 14}$ at RT

Prob: Ionic Product of water at 80°C is 2×10^{-12}. Calculate $(\log 2 = 0.3)$

① Neutral Pt

② Nature of soln (Acidic, Bacic or Neutral) if its pH is 5.5 (A), 6.9 (B), 6.7 (B)

Soln: $K_w = 2 \times 10^{-12} = [H^+][OH^-] = [H^+]^2$

$[H^+] = \sqrt{2} \times 10^{-6}$ (neutral)

$pH = 6 - \frac{1}{2}\log 2 = 5.85$

Calculation of pH for diff type of Solution :-

Case-1 St. Acid / St. Base

① $[H_n A] = a_0 M$

$[H^+] = n a_0 M$

$pH = -\log(n a_0)$

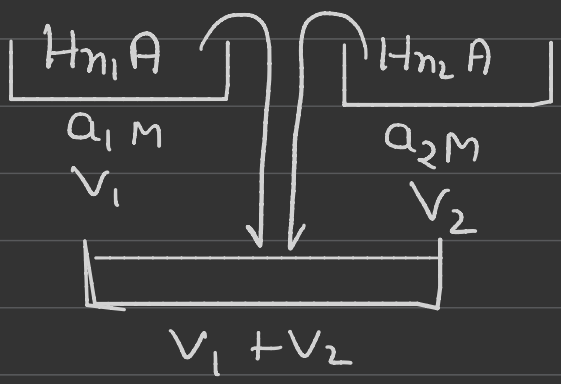
② $[B(OH)_n] = b_0 M$

$[OH^-] = n b_0 M$

$pOH = -\log(n b_0)$

$pH = 14 + \log(n b_0)$

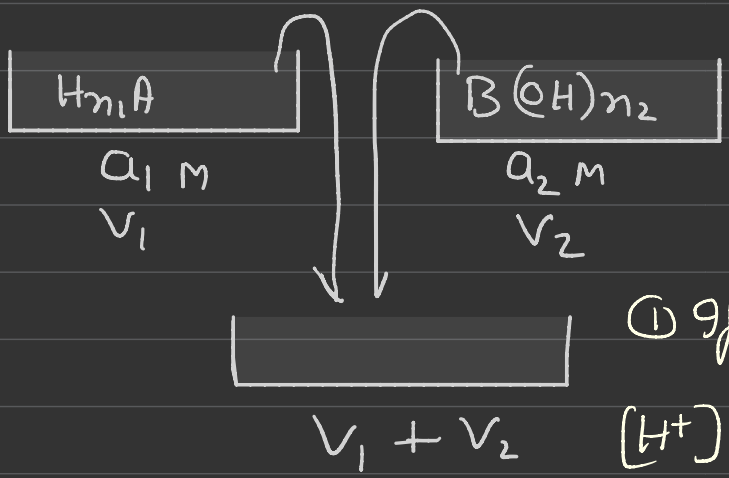
Case-2 Mix of St. Acids (or Mix of St. Bases)



$$[H^+] = \frac{n_1a_1V_1 + n_2a_2V_2}{V_1+V_2}$$

$$-\log[H^+] = pH$$

Case-3 Mix of (St. Acids + St. Bases)



$$n_{H^+} = n_1a_1V_1$$

$$n_{OH^-} = n_2a_2V_2$$

(i) If $n_{H^+} > n_{OH^-}$

$$[H^+] = \frac{n_1a_1V_1 - n_2a_2V_2}{V_1+V_2}$$

(ii) If $n_{H^+} < n_{OH^-}$

$$[OH^-] = \frac{n_2a_2V_2 - n_1a_1V_1}{V_1+V_2}$$

(iii) If $n_{H^+} = n_{OH^-}$ $\Rightarrow$ neutral

pH = 7

(1) $[H^+] = \frac{1\times0.01\times100}{400} = \frac{1}{400} \Rightarrow pH = 2.6$

(2) $[H^+] = \frac{2\times0.1\times20\phi}{50\phi} = \frac{4}{50}$ M $\Rightarrow$ pH = 1.1

(3) $[H^+] = \frac{1\times0.1\times100 + 1\&0.01\times1000}{1100} = \frac{10+10}{1100} = \frac{2}{110}$ M

pH = 1.75

Prob: Calculate pH ($\log 2 = 0.3$, $\log 5 = 0.7$, $\log 11 = 1.05$)

① 100 ml, 0.01 M HCl + 300 ml H_2O

② 200 ml, 0.1 M H_2SO_4 + 300 ml, 0.2 M Na^+Cl^- (× ×)

③ 100 ml, decimolar HCl + 1000 ml, centimolar HCl

Ans: ① pH = 2.6 ② pH = 1.1 ③ pH = 1.75

Case-4 Sol^n of Weak monoprotic Acid (or weak monohydroxy Base)

ⓐ HA = Weak monoprotic Acid

K_a = Ionization Cont

$$HA \rightleftharpoons H^+ + A^-$$

a_0

$a_0(1-\alpha)$ $\quad a_0\alpha \quad a_0\alpha$

$$K_a = \frac{(a_0\alpha)^2}{a_0(1-\alpha)} = \frac{a_0\alpha^2}{(1-\alpha)}$$

If $\frac{K_a}{a_0} \le 10^{-3}$ ✓

$(1-\alpha) \simeq 1$

$K_a = a_0\alpha^2$

$\alpha = \sqrt{\frac{K_a}{a_0}}$

$[H^+] = a_0\alpha = \sqrt{K_a a_0}$

If $\frac{K_a}{a_0} \le 10^{-3}$ ×

$K_a = \frac{a_0\alpha^2}{(1-\alpha)} \Rightarrow a_0\alpha^2 + K_a\alpha - K_a = 0$

ⓑ BOH = Weak monohydroxy base

K_b = Ionization Cont

Electrolytes :-

① **Strong Electrolytes :-** 100% ionizable

eg: St. Acid :- HCl, HBr, HI, H_2SO_4, $HClO_4$, $HBrO_4$, HIO_4, HNO_3 etc

St. Base :- hydroxide of alkali metal and alkaline earth metal.

Salts :-

② **Weak Electrolyte :-** Partially ionizable

eg: W. Acid :- HF, HNO_2, H_3PO_2, H_3PO_3, H_3PO_4, H_2S, CH_3COOH, $HCOOH$ etc

Organic Acids generally behave as weak Acids.

eg: W. Base :- NH_3 or NH_4OH, $Zn(OH)_2$, C_5H_5N, CH_3NH_2, $C_6H_5NH_2$ etc

Hydroxides of d-block generally behave as w.base

Organic bases are generally w. bases.

Strong electrolytes are more conducting than weak electrolyte in solution.

Ostwald's Dilution Law :-

A^+B^- = Binary weak Electrolyte

K = Ionization Cont

$$AB_{aq} \overset{K}{\rightleftharpoons} A^+_{aq} + B^-_{aq}$$

$t=0 \quad a_0$

t_{eq}	$a_0(1-\alpha)$	$a_0\alpha$	$a_0\alpha$

$$K = \frac{a_0\alpha^2}{(1-\alpha)}$$

For very weak electrolyte

$(1-\alpha) \simeq 1$

$$K = a_0\alpha^2$$

$$\boxed{\alpha = \sqrt{\frac{K}{a_0}}}$$

$$\Rightarrow \boxed{\alpha \propto \sqrt{V}}$$

$$\Rightarrow \boxed{\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{V_1}{V_2}}}$$

$\Rightarrow V \to \infty \Rightarrow \alpha \simeq 1$

Note: If $\boxed{\frac{K}{a_0} \leq 10^{-3}}^{*}$

$\Rightarrow (1-\alpha) \simeq 1$

Theory of Acids and Bases :-

① **Arrhenius theory :-** Substance which provide H^+ to soln are Acids while which provide OH^- to soln are base.

eg: $HClO_{4\,aq} \longrightarrow H^+ + ClO_4^-$ (Acid)

$CsOH_{aq} \longrightarrow OH^- + Cs^+$ (Base)

② **Bronsted Lowry theory :-** Substance which provide H^+ to soln are Acids.

Substance which take H^+ from soln are base.

eg: $CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$

Acid Base Base Acid

CH_3COOH / CH_3COO^- Conjugate Acid/Base

H_2O / H_3O^+ Conjugate Base / Acid

① eg:

Acid	Conjugate Base
HCl	Cl^-
H_2CO_3	HCO_3^-
H_3PO_4	$H_2PO_4^-$
$H_2PO_4^-$	HPO_4^{-2}
NH_4^+	NH_3

H_3PO_4 HPO_4^{-2} ✗

② eg: $HCl > HF$ (given) $\Rightarrow Cl^- < F^-$

$HCOOH > CH_3COOH$ (given) $\Rightarrow HCOO^- < CH_3COO^-$

$C_2H_4 < C_2H_2$ (given) $\Rightarrow C_2H_3^- > C_2H^-$

③ eg:

HCl (SA) Cl^- ✗

eg: $HClO_4$ (S.A) ClO_4^- ✗

eg: $NaOH$ (S.B) Na^+ ✗

Na^+Cl^- Neutral ✗ ✗

$CH_3COO^-Na^+$ Basic ✓ ✗

$NH_4^+Cl^-$ Acidic ✓ ✗

$ZnCl_2$ Acidic ✓ ✗

④ (a) CH_3COONa ✓ ✗

$$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$$

(b) NH_4Cl ✓ ✗

$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

(c) $H_3PO_3 \rightleftharpoons H^+ + H_2PO_3^-$

$H_2PO_3^- \rightleftharpoons H^+ + HPO_3^{--}$

(d) $Zn(OH)_2 \rightleftharpoons Zn^+(OH) + OH^-$

$Zn^+(OH) \rightleftharpoons Zn^{+2} + OH^-$

(e) Na_2CO_3

X ✓

$CO_3^{--} + H_2O \rightleftharpoons HCO_3^- + OH^-$

$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$

(3) Lewis Acid/Base theory :-

Acid : accept e⁻ pair
Base : donate e⁻ pair

Acid :-

(a) +ve charge
(b) incomplete octate
(c) Vacant d-orbital
(d) ↓

eg: CH_3^+

BH_3
↑

I_2
↑

$O=\overset{+\delta}{C}=O^{-\delta}$
↑

Base :

(a) -ve charge
(b) lone pair
(c)

eg: CH_3^-, OH^-, I^-

$:NH_3$, $C_5H_5N:$
↑

$(C_5H_5)_2Fe$ (Ferrocene)

Ionic Product of water: (K_w)

$$H_2O \underset{}{\overset{K_{eq}}{\rightleftharpoons}} H^+ + OH^- \quad : \Delta H = 57.1\ KJ \text{ or } 13.7\ KCal$$

a_0

$a_0(1-\alpha) \simeq a_0 \qquad a_0\alpha \qquad a_0\alpha$

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]} \Rightarrow \boxed{[H^+][OH^-] = K_w = 10^{-14}} \text{ at RT}$$

= Ionic Product of water

$$= \frac{10^{-14}}{\left(\frac{1000}{18}\right)} = 1.8 \times 10^{-16}$$

$$K_{eq} = 1.8 \times 10^{-16} = \frac{(a_0\alpha)^2}{a_0(1-\alpha)} = a_0\alpha^2$$

$$\alpha^2 = \frac{1.8 \times 10^{-16}}{\left(\frac{1000}{18}\right)} = (1.8)^2 \times 10^{-18}$$

$$\alpha = 1.8 \times 10^{-9}$$

Note: ① $T\uparrow \Rightarrow \xrightarrow{Rxn} \Rightarrow H^+\uparrow$ & $OH^-\uparrow \Rightarrow K_w\uparrow$

② for H_2O

$$[H^+] = [OH^-]$$

$$[H^+][H^+] = 10^{-14} \Rightarrow [H^+] = 10^{-7}\ M = [OH^-]$$

If $[H^+] > [OH^-] \Rightarrow [H^+] > 10^{-7}\ M \Rightarrow$ Acidic

$[H^+] < [OH^-] \Rightarrow [H^+] < 10^{-7}\ M \Rightarrow$ Basic

$[H^+] = [OH^-] \Rightarrow [H^+] = 10^{-7}\ M \Rightarrow$ Neutral

③

Relation b/w Ka and K_b of Conjugate Acid/Base :-

HA = Weak Acid , Ka = Ionization Cont
A^- = Conjugate Base , K_b = Ionization Const

$$HA \underset{}{\overset{K_a}{\rightleftharpoons}} H^+ + A^-$$

$$K_a = \frac{[H^+][A^-]}{[HA]} \quad -(I)$$

$$A^- + H_2O \overset{K_b}{\rightleftharpoons} HA + OH^-$$

$$K_b = \frac{[HA][OH^-]}{[A^-]} \quad -(II)$$

Eqn (I) × (II) *

$$K_a \times K_b = [H^+][OH^-] = K_w = 10^{-14} \text{ at RT}$$

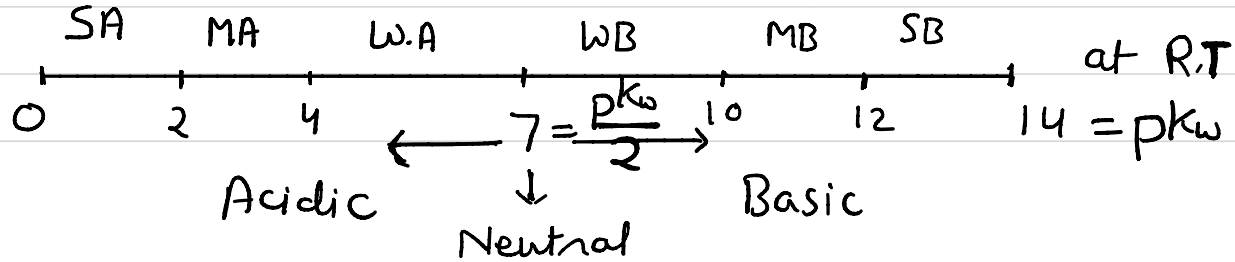
eg: CH_3COOH $(K_a = 1.8 \times 10^{-5})$

CH_3COO^- $K_b = \frac{10^{-14}}{1.8 \times 10^{-5}} = \left(\frac{10}{1.8}\right) \times 10^{-10}$

pH Scale :-

① $pH = -\log[H^+]$
$pOH = -\log[OH^-]$

②



③ $pK = -\log K$

$pK_a = -\log K_a$

$pK_b = -\log K_b$

$pK_w = -\log K_w = 14$ at RT

④ $K_a \times K_b = K_w$

$\boxed{pK_a + pK_b = pK_w}$

⑤ $[H^+][OH^-] = K_w$

$\boxed{pH + pOH = pK_w = 14}$ at RT

Prob: Ionic Product of water at 80°C is 2×10^{-12}. Calculate $(\log 2 = 0.3)$

① Neutral Pt

② Nature of soln (Acidic, Bacic or Neutral) if its pH is 5.5 (A), 6.9 (B), 6.7 (B)

Soln: $K_w = 2 \times 10^{-12} = [H^+][OH^-] = [H^+]^2$

$[H^+] = \sqrt{2} \times 10^{-6}$ neutral

$pH = 6 - \frac{1}{2}\log 2 =$ 5.85

Calculation of pH for diff type of Solution :-

Case-1 St. Acid / St. Base

① $[H_nA] = a_0 M$

$[H^+] = n a_0 M$

$pH = -\log(n a_0)$

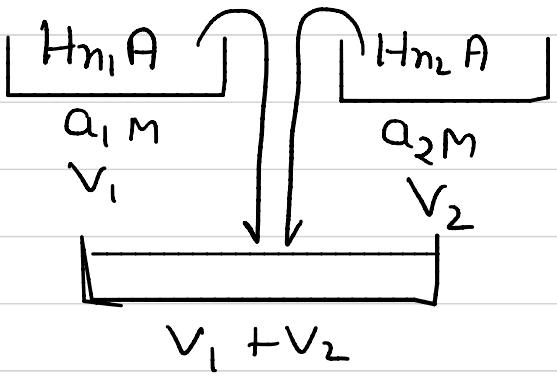
② $[B(OH)_n] = b_0 M$

$[OH^-] = n b_0 M$

$pOH = -\log(n b_0)$

$pH = 14 + \log(n b_0)$

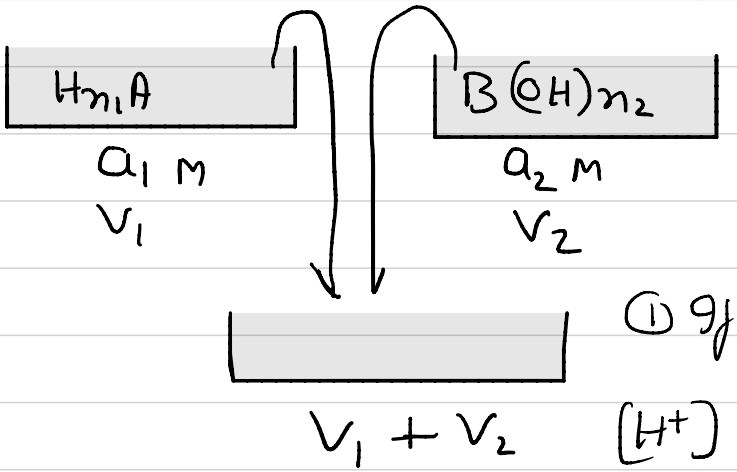
Case-2 Mix of St. Acids (or Mix of St. Bases)



$$[H^+] = \frac{n_1 a_1 V_1 + n_2 a_2 V_2}{V_1 + V_2}$$

$$-\log [H^+] = pH$$

Case-3 Mix of (St. Acids + St. Bases)



$$n_{H^+} = n_1 a_1 V_1$$

$$n_{OH^-} = n_2 a_2 V_2$$

(i) If $n_{H^+} > n_{OH^-}$

$$[H^+] = \frac{n_1 a_1 V_1 - n_2 a_2 V_2}{V_1 + V_2}$$

(ii) If $n_{H^+} < n_{OH^-}$

$$[OH^-] = \frac{n_2 a_2 V_2 - n_1 a_1 V_1}{V_1 + V_2}$$

(iii) If $n_{H^+} = n_{OH^-} \Rightarrow$ neutral

pH = 7

(1) $$[H^+] = \frac{1 \times 0.01 \times 100}{400} = \frac{1}{400} \Rightarrow pH = 2.6$$

(2) $$[H^+] = \frac{2 \times 0.1 \times 20\cancel{0}}{50\cancel{0}} = \frac{4}{50} M \Rightarrow pH = 1.1$$

(3) $$[H^+] = \frac{1 \times 0.1 \times 100 + 1 \times 0.01 \times 1000}{1100} = \frac{10+10}{1100} = \frac{2}{110} M$$

pH = 1.75

Prob: Calculate pH ($\log 2 = 0.3$, $\log 5 = 0.7$, $\log 11 = 1.05$)

① 100 ml, 0.01 M HCl + 300 ml H_2O

② 200 ml, 0.1 M H_2SO_4 + 300 ml, 0.2 M Na^+Cl^- ✗✗

③ 100 ml, decimolar HCl + 1000 ml, centimolar HCl

Ans: ① pH = 2.6 ② pH = 1.1 ③ pH = 1.75

Case-4 Sol^n of Weak monoprotic Acid (or weak monohydroxy Base)

ⓐ HA = Weak monoprotic Acid

K_a = Ionization Cont

$$HA \rightleftharpoons H^+ + A^-$$

a_0

$a_0(1-\alpha)$ $\quad a_0\alpha \quad a_0\alpha$

$$K_a = \frac{(a_0\alpha)^2}{a_0(1-\alpha)} = \frac{a_0\alpha^2}{(1-\alpha)}$$

If $\frac{K_a}{a_0} \le 10^{-3}$ ✓

$(1-\alpha) \simeq 1$

$K_a = a_0\alpha^2$

$$\alpha = \sqrt{\frac{K_a}{a_0}}$$

$$[H^+] = a_0\alpha = \sqrt{K_a a_0}$$

If $\frac{K_a}{a_0} \le 10^{-3}$ ✗

$$K_a = \frac{a_0\alpha^2}{(1-\alpha)} \Rightarrow a_0\alpha^2 + K_a\alpha - K_a = 0$$

ⓑ BOH = Weak monohydroxy base

K_b = Ionization Cont

How much water must added to 300 ml of 0.2 M solution of CH_3COOH for the degree of dissociation of the acid to double? K_a for the acetic acid = 1.8×10^{-5}.

(a) 1200 ml (b) 300 ml
(c) 600 ml (d) 900 ml

D

$\alpha \propto \sqrt{V}$

$\frac{\alpha_1}{\alpha_2} = \sqrt{\frac{V_1}{V_2}} \Rightarrow \frac{1}{2} = \sqrt{\frac{300}{300+V}}$

$\Rightarrow \frac{1}{4} = \frac{300}{300+V}$

$V = 900$ ml

What will be the resultant pH when 200 ml of an aqueous solution of HCl (pH = 2.0) is mixed with 300 ml of an aqueous solution of NaOH (pH = 12.0)?

(a) 2.7 (b) 11.3
(c) 3.7 (d) 10.3

B

$HCl \Rightarrow [H^+] = 10^{-2}$ M, 200 ml

$n_{H^+} = 2$ mmol

$NaOH \Rightarrow [OH^-] = 10^{-2}$ M, 300 ml

$n_{OH^-} = 3$ mmol

$[OH^-] = \frac{3-2}{500} = \frac{2}{1000}$

$pOH = 3 - \log 2 \Rightarrow pH = 14 - (3 - \log 2)$

$pH = 11.3$

What is the pH of 6.67×10^{-3} M aqueous solution of $Al(OH)_3$ if its first dissociation is 100%, second dissociation is 50% and the third dissociation is negligible.

(a) 2 (b) 12
(c) 11 (d) 3

B

$Al(OH)_3 \xrightarrow{100\%} Al(OH)_2^+ + OH^-$

$a_0 \qquad a_0 \qquad a_0$

$Al(OH)_2^+ \xrightarrow{50\%} Al(OH)^{+2} + OH^-$

$a_0 \qquad a_0/2 \qquad a_0/2$

$[OH^-] = \frac{3}{2}a_0 = \frac{3}{2} \times \frac{16}{3} \times 10^{-3} = 10^{-2}$

$pOH = 2 \Rightarrow pH = 12$

Dissociation constants of acids HA and HB are 2.0×10^{-4} and 5×10^{-5}, respectively. The $[H^+]$ in the resulting solution obtained by mixing 20 ml of 0.5 M – HA solution and 30 ml of 0.2 M – HB solution is

(a) 1.05×10^{-2} M (b) 6.78×10^{-3} M
(c) 1.05×10^{-3} M (d) 6.78×10^{-2} M

B

$[HA] = \frac{20 \times 0.5}{50} = 0.2$ M

$[HB] = \frac{30 \times 0.2}{50} = 0.12$ M

$[H^+] = \sqrt{2 \times 10^{-4} \times 0.2 + 5 \times 10^{-5} \times 0.12}$

$= \sqrt{4 \times 10^{-5} + 0.6 \times 10^{-5}}$

$= \sqrt{46 \times 10^{-6}} = 6.78 \times 10^{-3}$

H_3A is a weak tribasic acid with $K_{a1} = 10^{-5}$, $K_{a2} = 10^{-9}$ and $K_{a3} = 10^{-13}$. The value of pX of 0.1 M – H_3A solution, where $pX = -\log_{10}X$ and $X = \frac{[A^{3-}]}{[HA^{2-}]}$, is

(a) 5.0 (b) 4.0
(c) 9.0 (d) 10.0

D

$\left(\frac{z}{y}\right)$

$K_{a_1} = \frac{x^2}{a_0 - x} \Rightarrow \frac{K_{a_1}}{a_0} \leq 10^{-3}$

$x = \sqrt{K_{a_1} a_0} = 10^{-3}$ M

$K_{a_2} = y$

$K_{a_3} = \frac{xz}{y} \Rightarrow \frac{z}{y} = \frac{10^{-13}}{10^{-3}} = 10^{-10}$

If $\frac{k_b}{a_0} \leq 10^{-3}$ ✓ $\Rightarrow [OH^-] = \sqrt{k_b a_0}$

If $\frac{k_b}{a_0} \leq 10^{-3}$ ✗ $\Rightarrow a_0\alpha^2 + k_b\alpha - k_b = 0$

Common Ion effect :-

AB = weak electrolyte

$$AB \rightleftharpoons A^+ + B^-$$

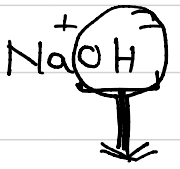
If $B^-\uparrow \Rightarrow \xleftarrow{Rxn} \Rightarrow \alpha\downarrow$

Where A^+ & B^- are called common ions

Odd ion effect :-

$$HA \rightleftharpoons H^+ + A^-$$

$[H^+]\downarrow \Rightarrow \longrightarrow \Rightarrow \alpha\uparrow$



odd ion

Case-5 Mix of weak Acid and St. Acid (or Mix of weak Base and St. Base)

HA + HCl
a_0 M b_0 M

$$HA \underset{}{\overset{K_a}{\rightleftharpoons}} H^+ + A^-$$

a_0		
$a_0(1-\alpha)$	$(b_0 + a_0\alpha)$	$a_0\alpha$

K_a = Ionization Cont

$$K_a = \frac{(b_0 + a_0\alpha)(a_0\alpha)}{a_0(1-\alpha)}$$

If $\frac{k_a}{a_0} \leq 10^{-3}$ ✓ $\Rightarrow K_a = \frac{(b_0 + a_0\alpha)\alpha}{1}$ — (1)

If $b_0 \geq a_0 \Rightarrow b_0 + a_0\alpha = b_0 \Rightarrow K_a = b_0\alpha$

$\Rightarrow [H^+] = b_0 + a_0\alpha \simeq \boxed{b_0}$ $\qquad \alpha = \left(\frac{K_a}{b_0}\right)$

Case-6 Mix of Weak Acids (or Mix of weak bases)

HA + HB

$a_1 \qquad a_2$ M

K_{a_1}, K_{a_2} = Ionization Cont

HA	$\rightleftharpoons$	H^+ +	A^-
a_1	K_{a_1}		
$(a_1 - x)$		$(x+y)$	x
HB	$\rightleftharpoons$	H^+ +	B^-
a_2	K_{a_2}		
$(a_0 - y)$		$(x+y)$	y

If $\left(\frac{K}{a}\right) \leq 10^{-3}$

$(a_1 - x) \simeq a_1$

$(a_2 - y) \simeq a_2$

$K_{a_1} = \frac{(x+y)\,x}{a_1}$

$(x+y)\,x = K_{a_1} a_1$ — (I)

$(x+y)\,y = K_{a_2} a_2$ — (II)

(I) + (II)

$(x+y)^2 = K_{a_1} a_1 + K_{a_2} a_2$

$$[H^+]\,(x+y) = \sqrt{K_{a_1} a_1 + K_{a_2} a_2}$$

$x = \frac{K_{a_1} a_1}{[H^+]}$

$y = \frac{K_{a_2} a_2}{[H^+]}$

Case-7 Polyprotic Weak Acid (or Polyhydroxy Weak Base)

H_3A = Triprotic weak Acid = a_0 M

$K_{a_1}, K_{a_2}, K_{a_3}$ = Ionization Cont

$\Rightarrow Ka_1 \gg Ka_2 \gg Ka_3$

$\Rightarrow x \gg y \gg z$

$x+y+z \simeq x$

$x-y \simeq x$

$y-z \simeq y$

$$H_3A \underset{}{\overset{Ka_1}{\rightleftharpoons}} H^+ + H_2A^-$$

$a_0 \quad (x+y+z) \quad (x-y)$

(a_0-x)

$$H_2A^- \overset{Ka_2}{\rightleftharpoons} H^+ + HA^{-2}$$

$x \quad (x+y+z) \quad (y-z)$

$(x-y)$

$$HA^{-2} \overset{Ka_3}{\rightleftharpoons} H^+ + A^{-3}$$

$y-z \quad (x+y+z) \quad z$

$[H^+] \simeq [H_2A^-] = x$

$[HA^{-2}] = y$

$[A^{-3}] = z$

$$\begin{cases} Ka_1 = \dfrac{x \times x}{a_0 - x} & -(1) \\ Ka_2 = \dfrac{x \times y}{x} \simeq y & -(11) \\ Ka_3 = \dfrac{x \times z}{y} & -(111) \end{cases}$$

$\Rightarrow$ If $\left(\dfrac{Ka_1}{a_0}\right) \leq 10^{-3}$ ✓

$(a_0 - x) \simeq a_0$

$[H^+] = x = \sqrt{Ka_1 a_0}$

Case-8 $[HCl] = 10^{-8}\,m = y$

$pH = 8$ ✗

$H^+ \leq 10^{-6}\,m$ or OH^-

$$H_2O \rightleftharpoons \underset{10^{-8}}{H^+} + \underset{(10^{-8}+x)}{OH^-}\ \ \underset{x}{}$$

$$[H^+][OH^-] = K_w = 10^{-14}$$

$$(10^{-8}+x)\,x = 10^{-14}$$

$$x^2 + 10^{-8}x - 10^{-14} = 0$$

$$x = \frac{-10^{-8} + \sqrt{10^{-16} + 4\times 10^{-14}}}{2}$$

$$x = \frac{-10^{-8} + 10^{-8}\sqrt{401}}{2}$$

$$[H^+] = 10^{-8} + x = \frac{10^{-8}\left[1+\sqrt{401}\right]}{2} < 10^{-7}\,m$$

$\Rightarrow pH < 7$

trick $\Rightarrow$ $[H^+] = \frac{y + 10^{-7}}{2}$

$y < 10^{-6}\,m$

. An aqueous solution is prepared by dissolving 0.1 mole H_2CO_3 in sufficient water to get 100 ml solution at 25°C. For H_2CO_3, $K_{a1} = 4.0 \times 10^{-6}$ and $K_{a2} = 5.0 \times 10^{-11}$. The only incorrect equilibrium concentration is

(a) $[H^+] = 6.32 \times 10^{-4}$ M

(b) $[HCO_3^-] = 2 \times 10^{-3}$ M

(c) $[CO_3^{2-}] = 5 \times 10^{-11}$ M

(d) $[OH^-] = 5 \times 10^{-12}$ M

Q :- pH of a dilute solution of HCl is 6.7. Calculate molarity of HCl solution. [redacted]

$[HCl] = x$ M

$H_2O \rightleftharpoons H^+ + OH^-$

$x+y \quad y$

$(x+y) \times y = 10^{-14}$ -(1)

$x + y = 2 \times 10^{-7}$

$y = \frac{10^{-14}}{2 \times 10^{-7}} = 0.5 \times 10^{-7}$

$x = 1.5 \times 10^{-7} = [HCl]$

$-\log(x+y) = 6.7 = 7 - 0.3$

$= -\log 10^{-7} - \log 2 = -\log 2 \times 10^{-7}$

The self-ionization constant for pure formic acid, $K = [HCOOH_2^+][HCOO^-]$ has been estimated as 10^{-6} M^2 and the density of formic acid is 1.15 g/cm^3 at room temperature. If 'x %' of formic acid molecules in pure formic acid is converted to formate ions, then the value of '1000x' is

$pH = 7 + \frac{1}{2}[pKa + \log a_0]$

SB + WA → Stronger → pH ↓

Smaller ⇒ Ka ↑

$HW > HX > HY > HZ$

. Separate solutions of NaW, NaX, NaY and NaZ, each of concentrations 0.1 M, has pH 7.0, 9.0, 10.0 and 11.0, respectively, at 25°C. The strongest acid among these is

(a) NaW (b) NaX

(c) NaY (d) NaZ

Q :- What is the H^+ ion concentration of a 0.4 M solution of NH_4Cl. K_b (NH_4OH) = 2.0×10^{-5}

$\frac{K_h}{a_0} \leq 10^{-3}$ ✓

$[H^+] = \sqrt{K_{eq} a_0}$

Acid

$= \sqrt{\frac{10^{-14}}{2 \times 10^{-5}} \times 0.4} = \sqrt{2 \times 10^{-10}}$

$= \sqrt{2} \times 10^{-5}$ M

If pH of 0.001 M potassium propionate solution be 8.0, then the dissociation constant of propionic acid will be

(a) 10^{-3} (b) 10^{-2} (c) $10^{-2.5}$ (d) 10^{-5}

$kA = 0.001\ M$

$pH = 7 + \frac{1}{2}[pka + \log a_0]$

$8 = 7 + \frac{1}{2}[pka - 3] \Rightarrow pka = 5$

$ka = 10^{-5}$

Q:- 0.2 M solution of pyridinium chloride $C_2H_5NH_3^+Cl^-$ was found to have a pH of 3.3. What is K_b for , $C_2H_5NH_2$?

WB + SA

$pH = 7 - \frac{1}{2}[pK_b + \log a_0] \Rightarrow 3.3 = 7 - \frac{1}{2}[pK_b + \log 0.2]$

$(-\log k_b + \log 0.2) = (7 - 3.3) \times 2$

$\log \frac{0.2}{k_b} = \log\left(\frac{10^8}{4}\right)$

$\log\left(\frac{0.2}{k_b}\right) = (4 - 0.3) \times 2$

$= 2[\log 10^4 - \log 2]$

$= 2\log\left(\frac{10^4}{2}\right)$

$k_b = 0.8 \times 10^{-8} = 8 \times 10^{-9}$

Q:- Calculate the pH of 1.0×10^{-2} M sodium Acetate, CH_3COONa. [$K_a(CH_3COOH) = 2.0 \times 10^{-5}$]
Also Calculate percentage hydrolysis ?

SB + WA

$k_h = \left(\frac{k_w}{k_a}\right)$

$\alpha = \sqrt{\frac{k_h}{a_0}} = \sqrt{\frac{10^{-14}}{2 \times 10^{-5} \times 10^{-2}}} = \sqrt{5 \times 10^{-8}} = \sqrt{5} \times 10^{-4}$

$pH = 7 + \frac{1}{2}[5 - \log 2 + \log 10^{-2}] = 7 + \frac{1}{2}[2.7] = 8.35$ (1.35)

Q:- What will be the pH of 4×10^{-5} M A_2B solution which is assumed to be dissociated 75% in water. [Given : K_b (AOH) = 5×10^{-9}; H_2B is strong acid]

(A) 4 (B) 5 (C) 6 (D) 10

$\log 3 \simeq 0.48$

Salt hydrolysis :-

① Complete Salt :-

eg: $NaCl$, CH_3COONa, NH_4Cl, $ZnCl_2$
Na_2CO_3 etc

② Incomplete Salt :-

eg: $NaHCO_3$, NaH_2PO_4, Na_2HPO_4
$Zn(OH)Cl$ etc.

Hydrolysis of Complete Salt :

① Salt of monoprotic Acid and monohydroxy base

ⓐ Salts of SA and SB :-

$$\underset{(SA)}{HCl} + \underset{(SB)}{NaOH} \longrightarrow \overset{+}{Na}\overset{-}{Cl} + H_2O$$

$\overset{\times}{\longleftarrow}$ $\times$ $\times$

Neutral Sol^n $\Rightarrow$ pH = 7
hydrolysis ✗

ⓑ Salts of SA and WB :-

$$\underset{(SA)}{HCl} + \underset{(WB)}{BOH} \longrightarrow \overset{+}{B}\overset{-}{Cl} + H_2O$$

$\checkmark$ $\overset{\times}{\longleftarrow}$ $\checkmark$

$$BCl + H_2O \rightleftharpoons BOH + HCl$$

$$\overset{a_0}{B^+} + Cl^- + H_2O \rightleftharpoons BOH + H^+ + Cl^-$$

$B^+ + H_2O \rightleftharpoons \underbrace{BOH}_{\downarrow K_b \text{ (given)}} + H^+$ (Net Ionic Eqn)

$K_a = K_h = \frac{K_w}{K_b}$

a_0

$a_0(1-\alpha)$ $\quad a_0\alpha \quad a_0\alpha$

If $\left(\frac{K_h}{a_0}\right) \leq 10^{-3}$ ✓ $\quad (1-\alpha) \simeq 1$

$$\alpha = \sqrt{\frac{K_h}{a_0}} \Rightarrow [H^+] = \sqrt{K_h . a_0}$$

$$\Rightarrow -\log[H^+] = -\frac{1}{2}\left[\log K_h + \log a_0\right]$$

$$\Rightarrow pH = 7 - \frac{1}{2}\left[pK_b + \log a_0\right]$$

② Salt of WA and SB :-

$HA + NaOH \longrightarrow Na^+A^- + H_2O$

(WA) (SB) ✓ ← X ✓

$NaA + H_2O \rightleftharpoons NaOH + HA$

$a_0 M$

$Na^+ + A^- + H_2O \rightleftharpoons Na^+ + OH^- + HA$

$A^- + H_2O \rightleftharpoons \underbrace{HA}_{\downarrow K_a \text{ (given)}} + OH^-$

$K_b = K_h = \frac{K_w}{K_a}$

a_0

$a_0(1-\alpha)$ $\quad a_0\alpha \quad a_0\alpha$

If $\left(\frac{K_h}{a_0}\right) \leq 10^{-3}$ ✓ $\quad (1-\alpha) \simeq 1$

$$\alpha = \sqrt{\frac{K_h}{a_0}} \Rightarrow [OH^-] = \sqrt{K_h a_0}$$

$$[OH^-] = \sqrt{\frac{K_w}{K_a} a_0}$$

$$\Rightarrow pH = 7 + \frac{1}{2}\left[pK_a + \log a_0\right]$$

(4) Salt of WA and WB :

$$HA + BOH \longrightarrow B^+A^- + H_2O$$

(WA) (WB)

$$B^+A^- + H_2O \rightleftharpoons BOH + HA$$

a_0

Net ionic Eqn

$$B^+ + A^- + H_2O \xrightleftharpoons{K_h} BOH + HA$$

a_0 a_0

$a_0(1-\alpha)$ $a_0(1-\alpha)$ $a_0\alpha$ $a_0\alpha$

$$\# \quad K_h = \frac{[BOH][HA]}{[B^+][A^-]} \times \frac{[H^+][OH^-]}{[H^+][OH^-]}$$

$$K_h = \frac{[BOH]}{[B^+][OH^-]} \times \frac{[HA]}{[H^+][A^-]} \times [H^+][OH^-]$$

$$K_h = \frac{K_w}{K_a \times K_b}$$

$K_h = \frac{(a_0\alpha)^2}{[a_0(1-\alpha)]^2} = \frac{\alpha^2}{(1-\alpha)^2} \Rightarrow \frac{\alpha}{1-\alpha} = \sqrt{K_h}$

$$\Rightarrow \boxed{\alpha = \frac{\sqrt{K_h}}{1+\sqrt{K_h}}} = \text{Cont}$$

If $(1-\alpha) \simeq 1$

$$\boxed{\alpha = \sqrt{K_h}}$$

$HA \underset{}{\overset{K_a}{\rightleftharpoons}} H^+ + A^-$

$$K_a = \frac{[H^+][A^-]}{[HA]} = \frac{[H^+] \times a_0(1-\alpha)}{a_0\alpha}$$

$$[H^+] = K_a \times \left(\frac{\alpha}{1-\alpha}\right) = K_a\sqrt{K_h}$$

$$\boxed{[H^+] = \sqrt{\frac{K_w K_a}{K_b}}}$$

$$\boxed{pH = 7 + \frac{1}{2}\left[pK_a - pK_b\right]}$$

Note: ① If $K_a > K_b$

$pH < 7$

$\Rightarrow$ Acidic

K_a for (HA)

K_b for (BOH)

If $K_a < K_b$

$pH > 7$

$\Rightarrow$ Basic

If $K_a = K_b$

$pH = 7 \quad \Rightarrow$ neutral

eg: $K_a (CH_3COOH) = K_b (NH_4OH) = 1.8 \times 10^{-5}$

$CH_3COONH_4 \Rightarrow$ Neutral $\Rightarrow pH = 7$

(2)

#

Salt	α	$[H^+]$ or $[OH^-]$	pH	K_h
SA + SB	—	$[H^+] = 10^{-7}$ M	7	—
SA + WB	$\sqrt{\frac{K_h}{a_0}}$	$[H^+] = \sqrt{K_h a_0}$	$7 - \frac{1}{2}[pK_b + \log a_0]$	$\left(\frac{K_w}{K_b}\right)$
WA + SB	$\sqrt{\frac{K_h}{a_0}}$	$[OH^-] = \sqrt{K_h a_0}$	$7 + \frac{1}{2}[pK_a + \log a_0]$	$\left(\frac{K_w}{K_a}\right)$
WA + WB	$\frac{\sqrt{K_h}}{1+\sqrt{K_h}}$	$[H^+] = \sqrt{\frac{K_w K_a}{K_b}}$	$7 + \frac{1}{2}[pK_a - pK_b]$	$\frac{K_w}{K_a . K_b}$

Prob: 100 ml, 0.4 M Sol^n of CH_3COOH is mixed with 100 ml, 0.4 M NaOH Sol^n. If K_a for CH_3COOH is 2×10^{-5}, then Calculate

(a) Neutralization cont for the rxn

. An aqueous solution is prepared by dissolving 0.1 mole H_2CO_3 in sufficient water to get 100 ml solution at 25°C. For H_2CO_3, $K_{a1} = 4.0 \times 10^{-6}$ and $K_{a2} = 5.0 \times 10^{-11}$. The only incorrect equilibrium concentration is

(a) $[H^+] = 6.32 \times 10^{-4}$ M

(b) $[HCO_3^-] = 2 \times 10^{-3}$ M

(c) $[CO_3^{2-}] = 5 \times 10^{-11}$ M

(d) $[OH^-] = 5 \times 10^{-12}$ M

Q :- pH of a dilute solution of HCl is 6.7. Calculate molarity of HCl solution. [redacted]

$[HCl] = x$ M

$H_2O \rightleftharpoons H^+ + OH^-$

$(x+y) \times (y) = 10^{-14}$ — ①

$-\log(x+y) = 6.7 = 7 - 0.3$

$= -\log 10^{-7} - \log 2$

$= -\log 2 \times 10^{-7}$

$x + y = 2 \times 10^{-7}$

$y = 0.5 \times 10^{-7}$, $x = 1.5 \times 10^{-7}$ M

The self-ionization constant for pure formic acid, $K = [HCOOH_2^+][HCOO^-]$ has been estimated as 10^{-6} M^2 and the density of formic acid is 1.15 g/cm^3 at room temperature. If 'x %' of formic acid molecules in pure formic acid is converted to formate ions, then the value of '1000x' is

. Separate solutions of NaW, NaX, NaY and NaZ, each of concentrations 0.1 M, has pH 7.0, 9.0, 10.0 and 11.0, respectively, at 25°C. The strongest acid among these is

(a) NaW ✓ (b) NaX

(c) NaY (d) NaZ

NaW + H_2O

NaOH + HW

HX HY HZ HW

HW > HX > HY > HZ

Q :- What is the H^+ ion concentration of a 0.4 M solution of NH_4Cl. K_b (NH_4OH) = 2.0×10^{-5}

$NH_4^+Cl^-$

✓ X

WB SA

$K_a = K_h = \left(\frac{K_w}{K_b}\right)$

If $\left(\frac{K_h}{c_0}\right) \leq 10^{-3}$ ✓

$[H^+] = \sqrt{\frac{10^{-14}}{2 \times 10^{-5}} \times 0.4}$

$[H^+] = \sqrt{\frac{2}{2}} \times 10^{-5}$ M

② Salts involving polyprotic Acid or polyhydroxy Base :-

$H_3A + 3NaOH \longrightarrow Na_3A + 3H_2O$

Weak triprotic Acid ✗ ✓

$\Rightarrow a_0M$

$A^{-3} + H_2O \underset{}{\overset{K_{b_1} \text{ or } K_{h_1}}{\rightleftharpoons}} HA^{-2} + OH^-$

$a_0 \quad (a_0-x) \quad x \quad (x+y+z)$

$HA^{-2} + H_2O \underset{}{\overset{K_{b_2} \text{ or } K_{h_2}}{\rightleftharpoons}} H_2A^- + OH^-$

$(x-y) \quad (y-z) \quad (x+y+z)$

$H_2A^- + H_2O \underset{}{\overset{K_{b_3} \text{ or } K_{h_3}}{\rightleftharpoons}} H_3A + OH^-$

$(y-z) \quad z \quad (x+y+z)$

$\Rightarrow$ given $K_{h_1} >> K_{h_2} >> K_{h_3}$

$x >> y >> z$

$$x+y+z \simeq x$$
$$x-y \simeq x$$
$$y-z \simeq y$$

$$K_{h_1} = \frac{x^2}{a_0-x} \quad -(i)$$

$$K_{h_2} = y \quad -(ii)$$

$$K_{h_3} = \frac{x \cdot z}{y} \quad -(iii)$$

given $K_{a_1}, K_{a_2}, K_{a_3}$ for H_3A

$$K_{h_3} = \left(\frac{K_w}{K_{a_1}}\right)$$

$$K_{h_2} = \left(\frac{K_w}{K_{a_2}}\right)$$

$$K_{h_1} = \left(\frac{K_w}{K_{a_3}}\right)$$

$[OH^-] = x$

If $\left(\frac{K_{h_1}}{a_0}\right) \leq 10^{-3}$ ✓

$$\boxed{[OH^-] = \sqrt{K_{h_1} a_0}}$$

② Hydrolysis of Incomplete Salt :-

eg: $H_2A + NaOH \rightleftharpoons \underset{\times}{Na^+HA^-} + H_2O$

K_{a_1} & K_{a_2} are given $\quad a_0\ M$

\# $HA^- \underset{(a_0-x-y)}{\overset{K_{a_2}}{\rightleftharpoons}} H^+ \underset{(x-z)}{} + A^{-2} \underset{x}{}$

$K_h = \left(\frac{K_w}{K_{a_1}}\right)$

$HA^- + H_2O \rightleftharpoons H_2A + OH^-$

$(a_0-x-y) \qquad y \qquad (y-z)$

$H^+ + OH^- \rightleftharpoons H_2O$

$x-z \qquad y-z$

$$\begin{cases} K_{a_2} = \dfrac{(x-z)\,x}{(a_0-x-y)} & -(I) \\ \dfrac{K_w}{K_{a_1}} = \dfrac{(y-z)\,y}{(a_0-x-y)} & -(II) \\ (x-z)(y-z) = K_w & -(III) \end{cases}$$

From eqn (I), (II), (III)

$$[H^+] = \sqrt{K_{a_1} K_{a_2}}$$

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2}$$

eg: $NaHCO_3$ $\Rightarrow$ $pH = \frac{pKa_1 + pKa_2}{2}$

eg: Na_2HPO_4 $\Rightarrow$ $pH = \frac{pKa_2 + pKa_3}{2}$

eg: NaH_2PO_4 $\Rightarrow$ $pH = \frac{pKa_1 + pKa_2}{2}$

eg: NaH_2PO_2 $\Rightarrow$ $pH = 7 + \frac{1}{2}[pKa + \log a_0]$

↓

Complete salt

Prob: 100 ml, 0.1 M Soln of H_2CO_3 is added separately with (i) 100 ml, 0.1 M NaOH & (ii) 200 ml, 0.1 NaOH

Calculate pH of soln. [$Ka_1 = 4 \times 10^{-7}$, $Ka_2 = 5 \times 10^{-10}$]

Soln: (i)

$H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

10 mmoh, 10 mmoh, 10 mmoh

$pH = \frac{pKa_1 + pKa_2}{2} = \frac{7 - \log 4 + 10 - \log 5}{2}$

$= \underline{7.85}$

(ii)

$H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

10, 20, 10 mmoh, 10 mmoh

$NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$

10, 10 mmoh, 10 mmoh

× ✓

$\Rightarrow a_0 = \frac{10}{300} = \frac{1}{30} M$

$1/30\ M$

$CO_3^{--} + H_2O \rightleftharpoons HCO_3^- + OH^-$

$\frac{1}{30} - x \qquad K_h = \left(\frac{K_w}{K_{a_2}}\right) \qquad x \qquad x$

$$K_h = \frac{10^{-14}}{5\times10^{-10}} = \frac{10^{-4}}{5}$$

q) $\frac{K_h}{C_0} \leq 10^{-3}$

$$\frac{10^{-4}}{5\times\frac{1}{30}} = \frac{3}{5}\times10^{-3} \leq 10^{-3} \checkmark$$

$$[OH^-] = \sqrt{\frac{10^{-4}}{5}\times\frac{1}{30}} = \sqrt{\frac{10^{-5}}{15}}$$

$$pOH = \frac{1}{2}\left[5 + \log 15\right] \Rightarrow pH = \underline{10.91}$$

Buffer Solution:-

① Mixed Buffer:

ⓐ Acidic Buffer Solution:-

eg: $HA + A^-$ $\Rightarrow$ $\frac{1}{10}$ to 10 mole ratio

WA C.B

eg: $CH_3COOH + CH_3COO^-$ ✓

$H_2CO_3 + HCO_3^-$ ✓

$HCN + CN^-$ ✓

$HCl + Cl^-$ ✗

pH Calculation :-

$HA \overset{K_a}{\rightleftharpoons} H^+ + A^-$

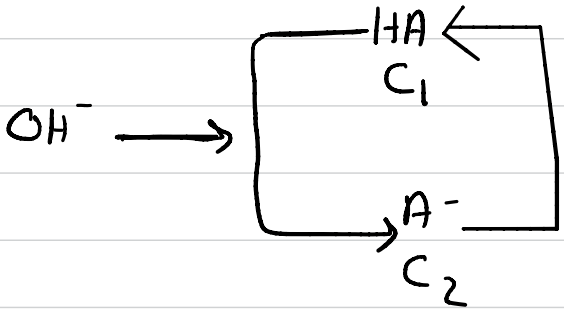
$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$[H^+] = K_a \times \frac{[HA]}{[A^-]} \Rightarrow \boxed{pH = pK_a + \log \frac{[A^-]}{[HA]}}$$

Handerson Eqn

Buffer action :-

eg: $HA + A^-$



$$pH_1 = pK_a + \log\left(\frac{C_2}{C_1}\right)$$

(a) Addition of Base :-

$$\underset{C_1}{HA} + \underset{C_0}{OH^-} \longrightarrow \underset{C_2}{A^-} + H_2O$$

$$(C_1 - C_0) \qquad (C_2 + C_0)$$

$$pH_2 = pK_a + \log\left(\frac{C_2 + C_0}{C_1 - C_0}\right)$$

(b) Addition of Acid :-

$$\underset{C_1}{HA} \longleftarrow \underset{C_2}{A^-} + \underset{C_0}{H^+}$$

(C_1+C_0) (C_2-C_0) —

$$pH_3 = pKa + \log\left(\frac{C_2-C_0}{C_1+C_0}\right)$$

Prob: Identify the buffer solution from the following.

(a) 0.5 mole CH_3COOH + 1 mole CH_3COONa ✗

(b) 1.5 mole CH_3COOH + 0.5 mole NaOH

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

1.5 mole 0.5 mole

1.0 mole — 0.5 mole

(c) 1.5 mole HCl + 0.5 mole NaOH

(d) 1.5 mole CH_3COONa + 1 mole HCl

$CH_3COONa + HCl \longrightarrow CH_3COOH + NaCl$

1.5 mole 1 mole

0.5 mole — 1 mole

(e) 2.5 mole H_2CO_3 + 0.5 mole NaOH

$H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

2.5 mole 0.5 mole

2.0 mole — 0.5 mole

(f) 2.5 mole H_2CO_3 + 3 mole NaOH

$H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

2.5 mole 3.0 mole

— 0.5 mole 2.5 mole

$NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$

2.5 0.5

2.0 mole — 0.5 mole

Blood

② Basic Buffer Solution:

eg: $BOH + B^+ \Rightarrow$ in range $\frac{1}{10}$ to 10

(WB) (C.A)

eg: ⓐ $NH_4OH + NH_4Cl$ ✓

ⓑ $C_6H_5NH_2 + C_6H_5NH_3^+Cl^-$ ✓

ⓒ $C_5H_5N + C_5H_5NH^+Cl^-$ ✓

ⓓ $NaOH + NaCl$ ✗

pH Calculation:-

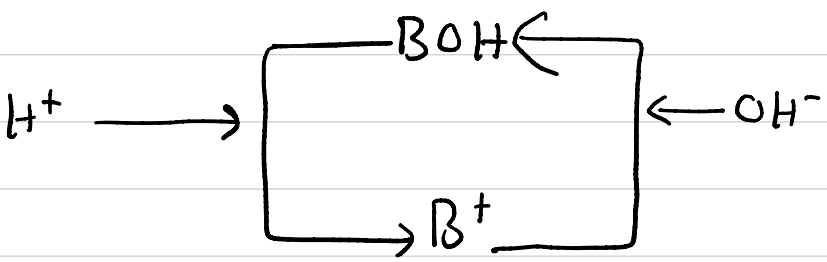
$$BOH \xrightleftharpoons{K_b} B^+ + OH^-$$

$$K_b = \frac{[B^+][OH^-]}{[BOH]}$$

$$[OH^-] = K_b \times \frac{[BOH]}{[B^+]}$$

$$pOH = pK_b + \log \frac{[B^+]}{[BOH]}$$

Buffer Action:-



#

Aniline behaves as a weak base. When 0.1 M, 50 mL solution of aniline was mixed with 0.1 M, 25 mL solution of HCl the pH of resulting solution was 8. Then the pH of 0.01 M solution of anilinium chloride will be ($K_w = 10^{-14}$) :

(a) 6 (b) 6.5

(c) 5 (d) 5.5

$C_6H_5NH_2 + HCl \longrightarrow C_6H_5NH_3Cl$

5 mmol 2.5 mmol — 2.5 mmol

2.5

$14 - 8 = pK_b + \log(1)$

$pK_b = 6$

$C_6H_5NH_3Cl = 0.01\ M$

$pH = 7 - \frac{1}{2}[6 + \log 10^{-2}]$

$= 5$

0.1 mole of CH_3NH_2 ($K_b = 5 \times 10^{-4}$) is mixed with 0.08 mole of HCl and diluted to 10 litre, what will be the OH^- concentration in the solution?

(a) 4.25×10^{-4} M (b) 1.25×10^{-4} M

(c) 7.9×10^{-4} M (d) 2.5×10^{-4} M

#

Which mixture(s) form(s) buffer solutions?

(P) 100 mL of 0.200 M HF + 200 mL of 0.200 M NaF

(Q) 200 mL of 0.200 M HCl + 200 mL of 0.400 M CH_3CO_2Na

(R) 300 mL of 0.100 M CH_3CO_2H + 100 mL of 0.300 M CH_3CO_2Na

(a) P only (b) R only

(c) Q and R only (d) P, Q and R

$k_a = 2 \times 10^{-5}$

$HA \rightleftharpoons H^+ + A^-$

(2 M) (1 M)

$A^- + H_2O \overset{K_h}{\rightleftharpoons} HA + OH^-$

1 2

#

100 mL solution (I) of buffer containing 0.1 (M) HA and 0.2 (M) A^-, is mixed with another solution (II) of 100 mL containing 0.2 (M) HA and 0.3 (M) A^-. After mixing, what is the pH of resulting solution?

[**Given :** pK_a of HA = 5]

(a) $5 - \log\frac{5}{3}$ (b) $5 + \log\frac{5}{3}$

(c) $5 + \log\frac{2}{5}$ (d) $5 - \log\frac{5}{2}$

$[HA] = \frac{10 + 20}{200} = \frac{3}{20}\ M$

$[A^-] = \frac{20 + 30}{200} = \frac{5}{20}\ M$

$pH = 5 + \log\left(\frac{5}{3}\right)$

(1) pH of 0.01 M – $(NH_4)_2SO_4$ and 0.02 M – NH_4OH buffer (pK_a of $NH_4^+ = 9.26$) is

(a) $9.26 + \log 2$ (b) $9.26 - \log 2$

(c) $4.74 + \log 2$ (d) 9.26

$pOH = (14 - 9.26) + \log\left(\frac{0.02}{0.02}\right)$

$pH = 9.26$

$NH_4^+ + NH_4OH$

$pH = 9.26 + \log\left(\frac{0.02}{0.02}\right) = 9.26$

(2) K_a for HCN is 5×10^{-10} at 25° C. For maintaining a constant pH of 9, the volume of 5 M KCN solution required to be added to 10 ml of 2 M HCN solution is (log 2 = 0.3) :

(a) 4 mL (b) 8 mL

(c) 2 mL (d) 10 mL

(2) $9 = 10 - \log 5 + \log\left(\frac{5V}{20}\right)$

$\log 5 + \log\frac{1}{10} = \log\left(\frac{V}{4}\right)$

$V = 2$ ml

(3) When 100 mL of 0.4 M CH_3COOH are mixed with 100 mL of 0.2 M NaOH, the $[H_3O^+]$ in the solution is approximately : [K_a (CH_3COOH) = 1.8×10^{-5}]

(a) 1.8×10^{-6} M (b) 1.8×10^{-5} M

(c) 9×10^{-6} M (d) 9×10^{-5} M

$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$

40 mmol 20 mmol 20 mmol

20

$[H^+] = Ka \times \frac{[CH_3COOH]}{[CH_3COO^-]}$

$= 1.8 \times 10^{-5}$ M

Which of the following solutions would have same pH?

(a) 100 mL of 0.2 M HCl + 100 mL of 0.4 M NH_3

(b) 50 mL of 0.1 M HCl + 50 mL of 0.2 M NH_3

(c) 100 mL of 0.3 M HCl + 100 mL of 0.6 M NH_3

(d) All will have same pH.

#

1 M benzoic acid ($pK_a = 4.20$) and 1 M C_6H_5COONa solutions are given separately. What is the volume of benzoic acid required to prepare a 300 ml buffer solution of pH = 4.5? **[log 2 = 0.3]**

(a) 200 mL
(b) 150 mL
(c) 100 mL
(d) 50 mL

V ml $\quad (300-V)$ ml

$C_6H_5COOH \quad C_6H_5COONa$

$4.5 = 4.2 + \log\frac{(300-V)}{V}$

$\log 2 = \log\left(\frac{300-V}{V}\right)$

$V = 100$ ml

What is the difference in pH for 1/3 and 2/3 stage of neutralisation of 0.1 M CH_3COOH with 0.1 M NaOH?

(a) $2\log 3$
(b) $2\log\left(\frac{1}{4}\right)$
(c) $2\log\left(\frac{2}{3}\right)$
(d) $2\log 2$

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

a_0

① $\frac{2a_0}{3} \quad \frac{a_0}{3}$

② $\frac{a_0}{3} \quad \frac{2a_0}{3}$

$pH_1 = pKa + \log\frac{a_0/3}{2a_0/3}$

$pH_2 = pka + \log\frac{2a_0/3}{a_0/3}$

$\Delta pH = 2\log 2$

To a solution of acetic acid, solid sodium acetate is gradually added. When 'x g' of the salt has been added, the pH has a certain value. When total 'y g' of the salt has been added, the pH has been further raised by 0.6 units. What is the ratio of $x:y$? (log 3.98 = 0.6)

(a) 3.98:1
(b) 1:3.98
(c) 2:3.98
(d) 3.98:2

$pH_1 = pka + \log\frac{(x/M)}{n}$

$pH_2 = pka + \log\frac{(y/M)}{n}$

$\Delta pH = 0.6 = \log\left(\frac{y}{x}\right) = \log(3.98)$

A volume of 10 ml of a strong acid solution of pH = 2.0 are mixed with 990 ml of a buffer solution of pH = 4.0. The pH of the resulting solution will be

(a) 4.2
(b) 6.0
(c) 4.002
(d) 4.0

$HA \quad A^- + H^+$

$C_1 \quad C_2 \quad 10^{-4}$

$C_1 + 10^{-4} \quad C_2 - 10^{-4}$

$pH = pka + \log\left(\frac{C_2}{C_1}\right) = 4$

$pH = pka + \log\left(\frac{C_2 - 10^{-4}}{C_1 + 10^{-4}}\right)$

$pH - 4 = \log\left[\frac{C_2 - 10^{-4}}{C_1 + 10^{-4}} \times \frac{C_1}{C_2}\right]$

Note:

① for Acidic Buffer [HA/A⁻]:-

$\frac{[A^-]}{[HA]} \Rightarrow$ ✗ $\frac{1}{10}$ ✓ 10 ✗

$pH \Rightarrow$ ✗ ✓ ✗

$(pK_a - 1)$ $(pK_a + 1)$

pH range from $(pK_a - 1)$ to $(pK_a + 1)$ is called working pH range of Buffer.

② Buffer Capacity :-

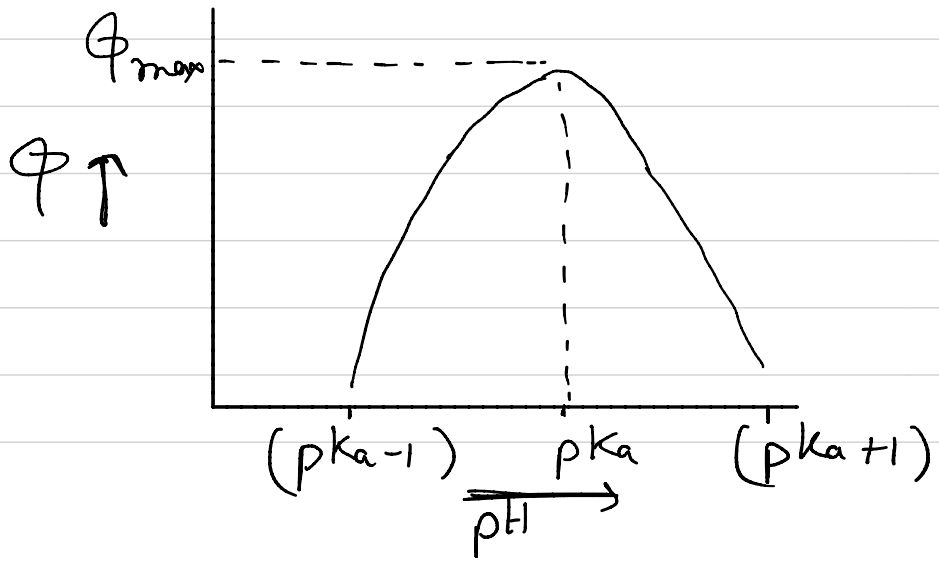
$$\text{Buffer index } (\phi) = \left[\frac{b}{d(pH)}\right]$$

where b = No of moles of Acid/Base added to 1 litre buffer soln

$d(pH)$ = Change in pH of soln

If $\frac{[A^-]}{[HA]} = 1 \Rightarrow pH = pK_a$

$\Rightarrow \phi = \phi_{max}$



#

$$pH = pKa + \log \frac{[A^-]}{[HA]}$$

$[A^-] = S$

$[HA] = a$

$[HA] + [A^-] = x$

$= \text{const}$

$$pH = pKa + \log\left(\frac{S}{x-S}\right) = pKa + \frac{1}{2.303} \times \ln \frac{(S)}{(x-S)}$$

$$\frac{d(pH)}{dS} = \frac{1}{2.303} \times \left[\frac{1}{S} + \frac{1}{x-S}\right]$$

where $dS = b$

$$\phi = \frac{dS}{d(pH)} = 2.303 \times \frac{S(x-S)}{x}$$

$$\frac{d\phi}{d(S)} = 0 = \frac{2.303}{x}\left[x - 2(S)\right]$$

$$S = \frac{x}{2} \Rightarrow 2(S) = a + (S)$$

$$\boxed{(S) = a}$$

$$[A^-] = [HA]$$

$$\Rightarrow (pH = pKa)$$

③ Effect of Dilution:-

eq: $HA \rightleftharpoons H^+ + A^-$

$t=0$ $\quad n_1 \qquad\qquad n_2$

$\left(\frac{n_1}{V_1}\right) \quad c \quad \left(\frac{n_2}{V_1}\right)$

If vol^m of Solution $= V_1$ lit

$$pH_1 = pKa + \log\left(\frac{n_2}{n_1}\right)$$

If Vol^m of Sol^n = V_2 lit

$$\frac{(n_1 - x)}{V_2} \qquad \left(\frac{cV_1 + x}{V_2}\right) \qquad \frac{(n_2 + x)}{V_2}$$

$$pH_2 = pK_a + \log\left(\frac{n_2 + x}{n_1 - x}\right)$$

$$K_a = \frac{\left(\frac{cV_1 + x}{V_2}\right) \times \left(\frac{n_2 + x}{V_2}\right)}{\left(\frac{n_1 - x}{V_2}\right)} = \frac{c \times \left(\frac{n_2}{V_1}\right)}{\left(\frac{n_1}{V_1}\right)}$$

$\Rightarrow$ x is very small

given $\begin{cases} K_a = 2 \times 10^{-5} \\ n_1 = n_2 = 0.5 \text{ mole} \\ V_1 = 1 \text{ lit} \\ V_2 = 100 \text{ lit} \end{cases}$

Indicators :-

$HIn \rightarrow$ Indicator

K_{In} = Ionization cont of Indicator

$$HIn \rightleftharpoons H^+ + In^-$$

Unionized form — Ionized form

Colour (A) — Colour (B)

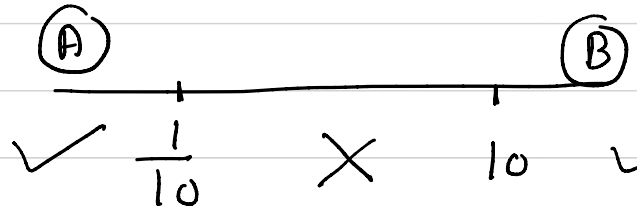
\# Acid $\Rightarrow H^+ \uparrow \Rightarrow \xleftarrow[\text{Rxn}]{} \Rightarrow HIn \uparrow$ & $In^- \downarrow \Rightarrow$ (A)

\# Base $\Rightarrow OH^- \uparrow \Rightarrow \xrightarrow[\text{Rxn}]{} \Rightarrow HIn \downarrow$ & $In^- \uparrow \Rightarrow$ (B)

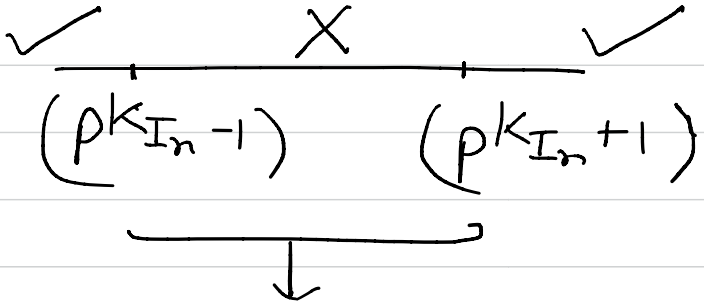
$$K_{In} = \frac{[H^+][In^-]}{[HIn]} \Rightarrow [H^+] = K_{In} \times \frac{[HIn]}{[In^-]}$$

$$\Rightarrow pH = pK_{In} + \log\frac{[In^-]}{[HIn]}$$

$$\Rightarrow \frac{[In^-]}{[HIn]} =$$



$pH \Rightarrow$



Colour transition Range

$$\% \text{ (colour A)} = \frac{[HIn]}{[HIn] + [In^-]} \times 100$$

How many grams of NaOH should be added in 500 ml of 2 M acetic acid solution to get a buffer solution of maximum buffer capacity?

(a) 20.0 (b) 10.0
(c) 40.0 (d) 30.0

For the indicator thymol blue, the value of pH is 2.0 when half of the indicator is present in the unionized form. The percentage of the indicator in the unionized form in a solution of 4.0×10^{-3} M hydrogen ion concentration is

(a) 40% (b) 28.6%
(c) 71.4% (d) 60%

$HI_n \rightleftharpoons H^+ + I_n^-$

$a_0 \quad \left(\frac{a_0}{2}\right) \quad \left(\frac{a_0}{2}\right)$

$pH = 2 = pK_{I_n} + \log(1)$

$[H^+] = K_{I_n} \times \left[\frac{HI_n}{I_n^-}\right]$

$4 \times 10^{-3} = 10^{-2} \times \frac{[HI_n]}{[I_n^-]}$

$\left(\frac{HI_n}{I_n^-}\right) = \frac{4}{10} \Rightarrow [HI_n]\% = \frac{4}{4+10} \times 100$

$= 28.6\%$

The range of most suitable indicator which should be used for titration of NaX (0.1 M, 10 ml) with 0.1 M HCl should be (K_b of X– = 10^{-6})

(a) 2–3 (b) 3–5
(c) 6–8 (d) 8–10

A certain indicator (an organic dye) has $pK_a = 5$. For which of the following titrations may it be suitable?

(a) Acetic acid against NaOH
(b) Aniline hydrochloride against NaOH
(c) Sodium carbonate against HCl
(d) Barium hydroxide against oxalic acid

An acid–base indicator has $K_a = 3.0 \times 10^{-5}$. The acid form of the indicator is red and the basic form is blue. The $[H^+]$ required to change the indicator from 75% blue to 75% red is

(a) 8×10^{-5} M
(b) 9×10^{-5} M
(c) 1×10^{-5} M
(d) 3.33×10^{-5} M

$[H^+]_2 - [H^+]_1 = (H^+)_{Req}$

	HIn	In^-
	R	B
①	25	75
②	75	25

$pH_1 = pK_{In} + \log \frac{3}{1}$

$pH_2 = pK_{In} + \log \frac{1}{3}$

$\Delta(pH) = 2 \log 3$

$[H^+] = 9 \times 10^{-5}$

$\Delta[H^+] = 8 \times 10^{-5}$

Titration :-

Equivalence Pt :

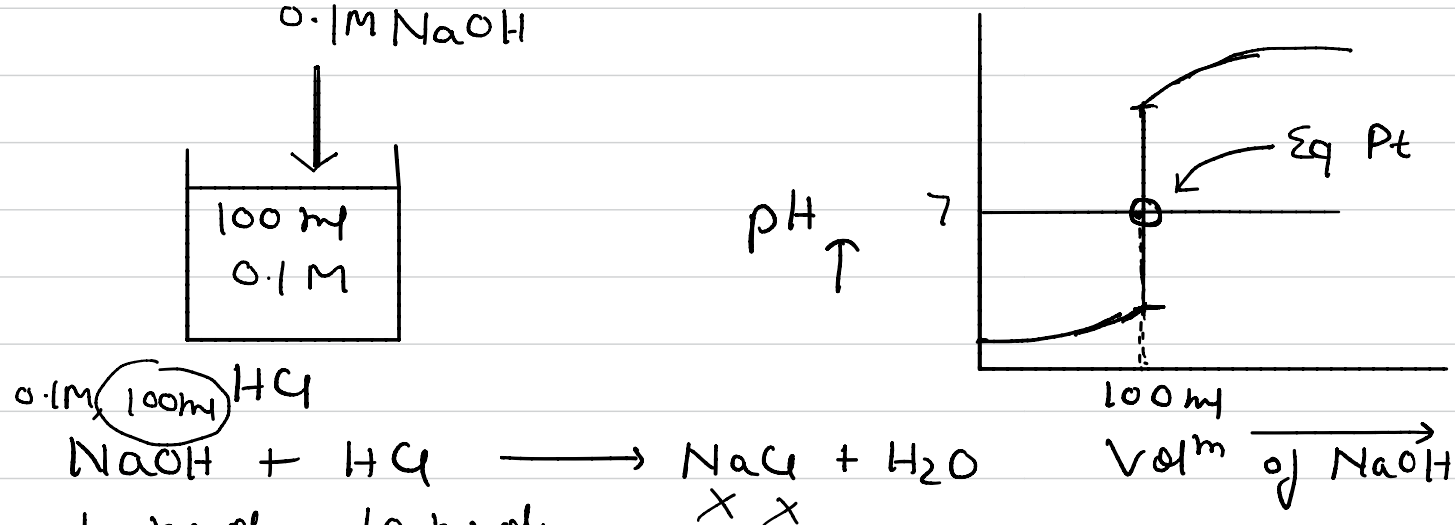
eg:

Acid / Base	No of Eq. Pt
HCl	1
H_2CO_3	2
H_3PO_4	3
CH_3COONa	1
$Zn(OH)_2$	2

End Point :

titration involving monoprotic Acids and monohydroxy Bases :

(1) Titration of St. Acid with St. Base :-



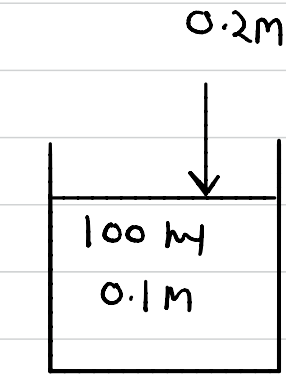
$NaOH + HCl \longrightarrow NaCl + H_2O$

10 mmole 10 mmole

— — 10 mmole

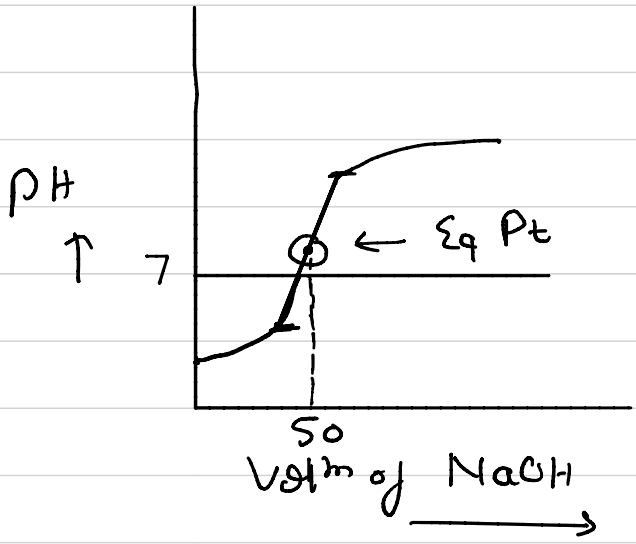
$\Rightarrow pH = 7$

② Titration of W. Acid with St. Base :-



$K_a = 1.8 \times 10^{-5}$

$pK_a = 4.74$



0.2M, (50 ml)

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

10 mmole 10 mmole

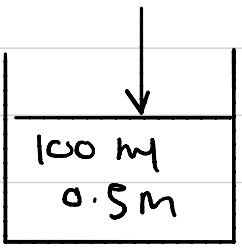
10 mmole

$$C_0 = \frac{10}{150} = \frac{1}{15} M$$

$$pH = 7 + \frac{1}{2}\left[4.74 - \log 15\right]$$

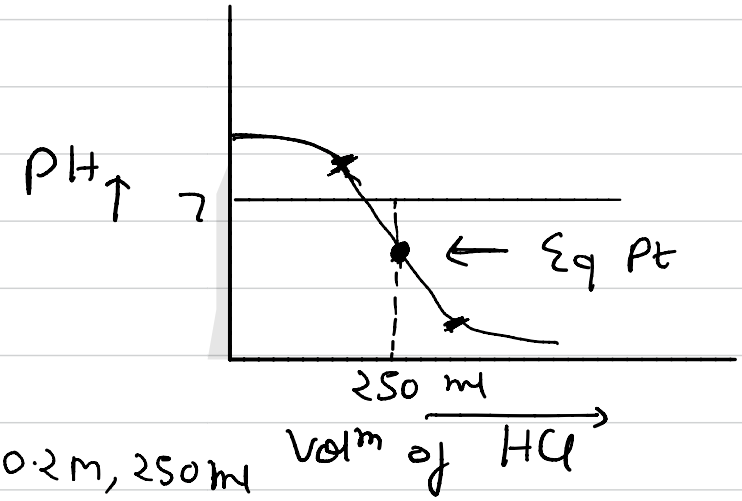
$$pH = 8.78$$

③ Titration of W Base with St. Acid :-



$K_b = 1.8 \times 10^{-5}$

$pK_b = 4.74$



0.2M, 250 ml

$$NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$$

50 50 mmole

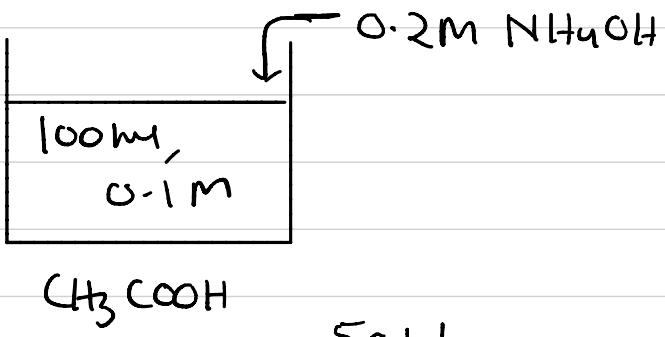
50 mmole

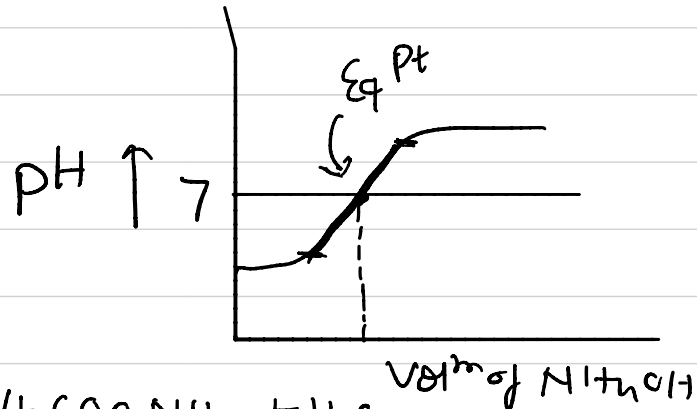
[given $\log 7 = 0.845$]

$$a_0 = \frac{5\phi}{35\phi} = \frac{1}{7}M$$

$$pH = 7 - \frac{1}{2}[4.74 - \log 7] = \underline{5.052}$$

(4) Titration of Weak Acid by W. base :-





50 ml

$$CH_3COOH + NH_4OH \longrightarrow CH_3COONH_4 + H_2O$$

10 mmole 10 mmole 10 mmole

$$pH = 7 + \frac{1}{2}[pK_a - pK_b]$$

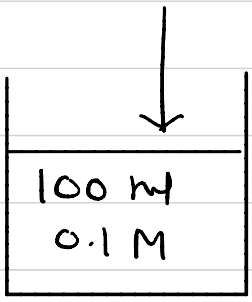
$$pH = 7$$

Note :-

Common Acid-Base Indicators

Indicator	Approximate pH Range for Color Change	Color Change
methyl orange	3.2 – 4.4	red to yellow
bromthymol blue	6.0 – 7.6	yellow to blue
phenolphthalein	8.2 – 10	colorless to pink
litmus	5.5 – 8.2	red to blue
bromcresol green	3.8 – 5.4	yellow to blue
thymol blue	8.0 – 9.6	yellow to blue

Titration involving polyprotic Acid / Polyhydroxy Base :-



given
$\begin{cases} \log 2 = 0.3 \\ \log 3 = 0.48 \end{cases}$

H_2CO_3

$\begin{cases} Ka_1 = 4 \times 10^{-7} \\ Ka_2 = 5 \times 10^{-11} \end{cases}$

	Vol^m of NaOH	pH	
①	0	3.7	
②	10 ml	5.8	
③	25 ml	6.4	
④	50 ml	8.35	Ist Eq Pt
⑤	80 ml	10.48	
⑥	100 ml	11.485	IInd Eq Pt

① H_2CO_3 0.1 M

If $\frac{Ka_1}{a_0} \leq 10^{-3}$ ✓

$[H^+] = \sqrt{Ka_1 a_0} = \sqrt{4 \times 10^{-7} \times 0.1}$

$= 2 \times 10^{-4}$ M

$pH = 4 - \log 2 = 3.7$

② $H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

10 mmoh, 2 mmoh → 8 mmoh, 2 mmoh

$pH = pKa_1 + \log\left(\frac{2}{8}\right) = 7 - \log 4 - \log 4$

$= 7 - 4 \times 0.3 = 5.8$

③ $H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

10 mmoh, 5 mmoh → 5 mmoh, 5 mmoh

$pH = pKa_1 \Rightarrow pH = 7 - \log 4 = 6.4$

④ $H_2CO_3 + NaOH \longrightarrow$ Salt $NaHCO_3 + H_2O$

10 mmol 10 mmol 10 mmol

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2} = \frac{7 - \log 4 + 11 - \log 5}{2}$$

$$= \frac{18 - \log 20}{2} = 9 - \frac{1}{2}\left[1.3\right]^{0.65} = 8.35$$

⑤ $H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

10 mmol 16 mmol

6 mmol 10 mmol

$NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$

10 mmol 6 mmol

4 mmol 6 mmol

$$pH = pK_{a_2} + \log \frac{6}{4} = 11 - \log 5 + \log 3 - \log 2$$

$$= 11 - 1 + 0.48 = 10.48$$

⑥ $H_2CO_3 + 2NaOH \longrightarrow Na_2CO_3 + 2H_2O$

(× Na_2CO_3, ✓ Na_2CO_3)

10 mmol 20 mmol

10 mmol

$$K_{h_1} = \frac{10^{-14}}{K_{a_2}}$$

$$a_0 = \frac{10}{20\phi} = \frac{1}{20} M$$

$CO_3^{--} + H_2O \rightleftharpoons HCO_3^- + OH^-$

$a_0 = \frac{1}{20} M$

$a_0 - x$ x x

$$\frac{K_{h_1}}{a_0} = \frac{10^{-14}}{5 \times 10^{-11}} \times 20 = 4 \times 10^{-3}$$

$$K_{h_1} = \left(\frac{x^2}{a_0 - x}\right)$$

$$x^2 + 2 \times 10^{-4} x - 10^{-5} = 0$$

$$x = \frac{-2 \times 10^{-4} + \sqrt{4 \times 10^{-8} + 4 \times 10^{-5}}}{2}$$

$$= -10^{-4} + \sqrt{10^{-5}}$$

$= -10^{-4} + 3.16 \times 10^{-3}$

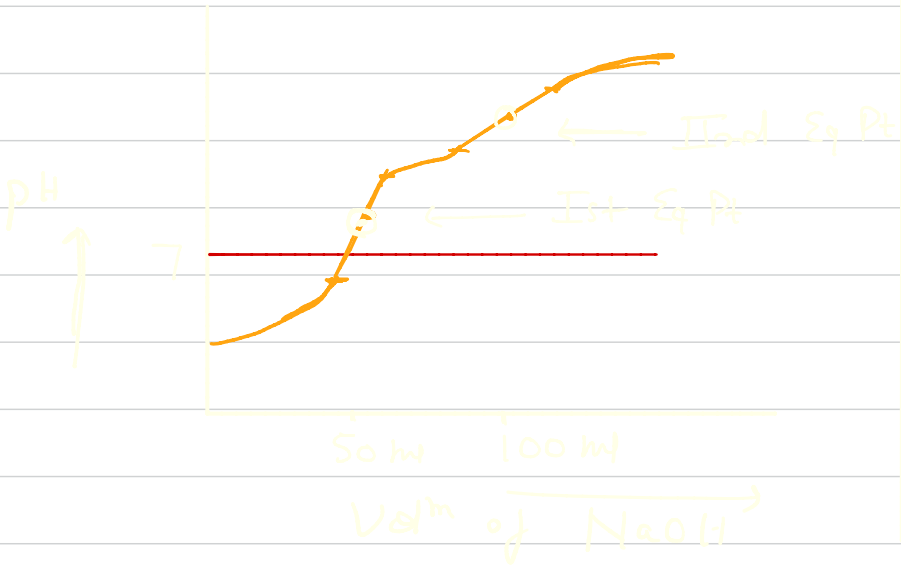
$= 30.6 \times 10^{-4}$

$x = [OH^-] = 30.6 \times 10^{-4}$

$pOH = 4 - \log 30.6 = 4 - 1.485$

$pH = 11.485$

Double indicator titration



titration of Na_2CO_3 with HCl using phenolphthalein indicator

titration of Na_2CO_3 with HCl using methyl orange indicator

Solubility (S):-

① Soluble Substance ⇒ (S) > 0.1 M
② Partially Soluble ⇒ 0.1M > (S) > 0.01 M
*③ Sparingly Soluble or Insoluble Substance ⇒ 0.01 M > (S)

Solubility Product (K_{sp}) :-

$$\underset{(S)}{A_xB_y(s)} + aq \rightleftharpoons A_xB_y(aq) \rightleftharpoons \underset{(S)}{x A^{+y}_{(aq)}} + \underset{s(1-\alpha)}{y B^{-x}_{(aq)}}$$

$x(S\alpha)$ $y(S\alpha)$

at Equib

$$\boxed{K_{sp} = [A^{+y}]^x \times [B^{-x}]^y}$$

$$K_{sp} = x^x (S\alpha)^x \times y^y (S\alpha)^y$$

$$\boxed{K_{sp} = x^x . y^y (S\alpha)^{x+y}}$$

Where
(S) = Solubility in pure water

If $\alpha = 1 \Rightarrow K_{sp} = x^x . y^y (S)^{x+y}$

eg: $AgCl_{(s)} \Rightarrow K_{sp} = (S)^2$
$Ag_2(CrO_4)_{(s)} \Rightarrow K_{sp} = 4(S)^3$
$A_2B_3(s) \Rightarrow K_{sp} = 108(S)^5$

Ionic Product (Q_{sp}):-

$$\underset{(S)}{A_xB_y(s)} \rightleftharpoons x A^{+y}_{(aq)} + y B^{-x}_{(aq)}$$

$$\boxed{Q_{sp} = [A^{+y}]^x \times [B^{-x}]^y}$$

at any instant

If $Q_{sp} > K_{sp} \Rightarrow \xleftarrow{Rxn} \Rightarrow$ PPt will form
or
Over saturated Soln

$Q_{sp} = K_{sp} \Rightarrow$ at Equib $\Rightarrow$ Saturated Soln
or
PPt formation will just Start.

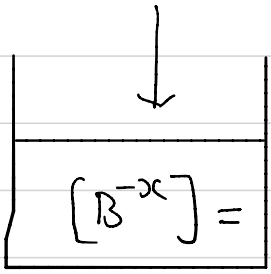
$Q_{sp} < K_{sp} \Rightarrow \xrightarrow{Rxn} \Rightarrow$ Unsaturated Soln

Note :- $A_xB_{y(s)} + aq \overset{K_{sp}}{\rightleftharpoons} x A^{+y}_{aq} + y B^{-x}_{aq}$

$$\ln \frac{(K_{sp})_2}{(K_{sp})_1} = \frac{-\Delta H}{R}\left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

Solubility in Presence of common ion :

A_xB_y



$A_xB_{y(s)} + aq \rightleftharpoons x A^{+y}_{aq} + y B^{-x}_{aq}$

(S') — $x(S')$ b_0 $b_0 + y(S') \simeq b_0$

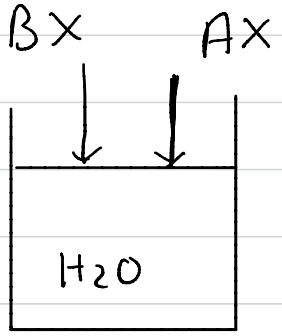
$$K_{sp} = [x(S')]^x \times b_0^y$$

$$S' = \frac{1}{x}\left[\frac{K_{sp}}{b_0^y}\right]^{1/x}$$

$b_0 \uparrow \Rightarrow S' \downarrow$

$\Rightarrow$ Presence of Common ion will decrease solubility.

Simultaneous Solubility :-



$$AX_{(s)} \underset{}{\overset{K_{sp_1}}{\rightleftharpoons}} A^+_{aq} + X^-_{aq}$$

S_1 — S_1 (S_1+S_2)

$$BX_{(s)} \overset{K_{sp_2}}{\rightleftharpoons} B^+_{aq} + X^-_{aq}$$

S_2 — S_2 (S_1+S_2)

$$S_1 \times (S_1+S_2) = K_{sp_1} \quad \text{—(I)}$$

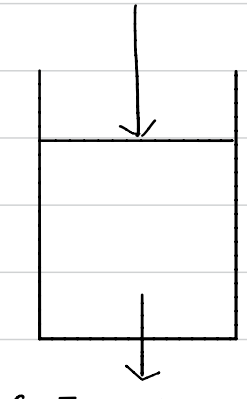
$$S_2 \times (S_1+S_2) = K_{sp_2} \quad \text{—(II)}$$

(I) + (II)

$$(S_1+S_2) = \sqrt{K_{sp_1} + K_{sp_2}}$$

$$S_1 = \frac{K_{sp_1}}{\sqrt{K_{sp_1} + K_{sp_2}}} \; ; \quad S_2 = \frac{K_{sp_2}}{\sqrt{K_{sp_1} + K_{sp_2}}}$$

Selective Precipitation :-



$K_{sp}(AgCl) = K_1$, $K_{sp}(AgBr) = K_2$, $K_{sp}(AgI) = K_3$

$$[Ag^+][Cl^-] = K_{sp} = K_1$$

$$[Ag^+]_1 = \frac{K_1}{[Cl^-]} \quad \text{-(I)}$$

For AgBr, $[Ag^+]_2 = \frac{K_2}{[Br^-]}$ —(II)

For AgI $[Ag^+]_3 = \frac{K_3}{[I^-]}$ —(III)

given $K_{sp}(AgI) < K_{sp}(AgBr) < K_{sp}(AgCl)$

If $[Cl^-] = [Br^-] = [I^-]$

$[Ag^+]_3 < [Ag^+]_2 < [Ag^+]_1$

AgI > AgBr > AgCl order of PPt

What is conc^n of I^- when precipitation of AgBr just start?

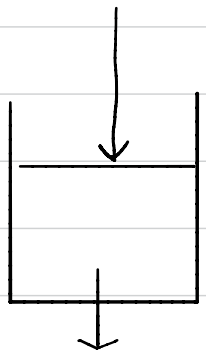
✓ $[Ag^+]_2 = \frac{K_2}{[Br^-]}$ for PPt of AgBr

$[Ag^+]_2 \times [I^-]_f = K_3$

$\Rightarrow [I^-]_f = \frac{K_3 \times [Br^-]}{K_2}$

Complex ion formation:-

AgCl



$[NH_3] = b_0\ M$

$AgCl_{(s)} \rightleftharpoons Ag^+_{aq} + Cl^-_{aq}$

(s) (s) (s)

$\simeq y$

$Ag^+_{aq} + 2\,NH_{3\,aq} \underset{K_f \simeq 10^8}{\rightleftharpoons} Ag(NH_3)_2^+{}_{aq}$

(s) b_0

$\simeq 0 \simeq y$ $[b_0 - 2(s)]$ (s)

$y \times (s) = K_{sp}(AgCl)$ — (i)

$\frac{(s)}{y \times [NH_3]^2} = K_f$ — (ii)

Eqn (i) × (ii) $\Rightarrow \frac{(s)^2}{[NH_3]^2} = K_f \times K_{sp}(AgCl)$

$(s) = [NH_3] \times \sqrt{K_f \times K_{sp}(AgCl)}$

$(S)_{pure} < (S)_{NH_3}$

Solubility of AgCN in Water :-

$$AgCN_{(s)} \rightleftharpoons Ag^+ + CN^-$$

(S)	S	S(1-α)

$$CN^- + H_2O \underset{K_h = \left(\frac{K_w}{K_a}\right)}{\rightleftharpoons} HCN + OH^-$$

S(1-α)		Sα	Sα

$$S \times S(1-\alpha) = K_{sp}(AgCN) \quad -(i)$$

$$\frac{(S\alpha)^2}{S(1-\alpha)} = K_h = \frac{K_w}{K_a} \quad -(ii)$$

Solubility of AgCN in Acidic Soln :-

$[H^+] = x\ M$

$$AgCN_{(s)} \rightleftharpoons Ag^+ + CN^-$$

(S)		(S)	y

$$CN^- + H^+ \rightleftharpoons HCN$$

S	x	
0 ≃ y	(x − S)	(S)

$$S \times y = K_{sp} \quad -(i)$$

$$\frac{y \times (x-S)}{(S)} = K_a \quad -(ii)$$

⇒ Where K_a is very Small

Similarly for solubility of AgCN in Base soln

If the solubility of lithium sodium hexafluorido aluminate, $Li_3Na_3(AlF_6)_2$ is 's' mol lt^{-1}, its solubility product is equal to :

(a) $729\ s^8$ (b) $12\ s^8$

(c) $3900\ s^8$ (d) $2916\ s^8$

$K_{sp} = 3^3 \times 3^3 \times 2^2 (S)^{3+3+2}$

$= 2916\ (S)^8$

In a saturated solution of Ag_2CO_3, silver ion concentration is 2×10^{-4} M. Its solubility product is :

(a) 4×10^{-12} (b) 3.2×10^{-11}

(c) 8×10^{-12} (d) 10^{-12}

$Ag_2CO_3(s) \rightleftharpoons 2Ag^+ + CO_3^{--}$

$(S) \qquad 2(S) \qquad (S)$

$K_{sp} = 4(S)^3 = 4 \times 10^{-12}$

$2(S) = 2 \times 10^{-4} \Rightarrow (S) = 10^{-4}$ M

. The solubility product of a salt having general formula MX_2, in water is 4×10^{-12}. The concentration of M^{2+} ions in the saturated aqueous solution of the salt is :

(a) 2.0×10^{-6} M (b) 1.0×10^{-4} M

(c) 1.6×10^{-4} M (d) 4.0×10^{-10} M

$MX_2(s) \rightleftharpoons M^{+2} + 2X^-$

$(S) \qquad S \qquad 2(S)$

$K_{sp} = 4(S)^3 = 4 \times 10^{-12}$

$(S) = 10^{-4}$ M $= [M^{+2}]$

A well is dug in a bed of rock containing fluorspar (CaF_2). If the well contains 20000 L of water, what is the amount of F^- in it?

$K_{sp} = 4 \times 10^{-11}$ $(10^{1/3} = 2.15)$

(a) 4.3 mol (b) 6.8 mol

(c) 8.6 mol (d) 13.6 mol

$K_{sp} = 4(S)^3 = 4 \times 10^{-11} \times 10 \times \frac{1}{10}$

$(S) = (10)^{1/3} \times 10^{-4}$ M

$= 2.15 \times 10^{-4}$ M

$2(S) = [F^-] = 4.30 \times 10^{-4} = \frac{n}{20000}$

$n = 8.6$ mole

The solubility product of AgCl is 10^{-10}. The minimum volume (in L) of water required to dissolve 1.722 mg of AgCl (molecular weight of AgCl = 143.5) :

(a) 10 lt (b) 2.2 lt

(c) 1.2 lt (d) 20 lt

$K_{sp} = (S)^2 = 10^{-10}$

$(S) = 10^{-5}$ M $= \frac{1.722 \times 10^{-3}}{143.5}$

$V_{(lit)}$

$V = \frac{172.2}{143.5} = 1.2$ lit

What is the pH of a saturated solution of magnesium hydroxide, $Mg(OH)_2$ at 25° C?
($K_{sp} = 6.0 \times 10^{-12}$ at 25° C)

(a) 10.56 (b) 10.36
(c) 10.26 (d) 10.05

$Mg(OH)_{2(s)} \rightleftharpoons Mg^{+2} + 2OH^-$

(s) s 2(s)

$K_{sp} = 6 \times 10^{-12} = 4(s)^3 \times 2$ (×2)

$[OH^-] = 2(s) = (12)^{1/3} \times 10^{-4}$ M

$pOH = 4 - \frac{1}{3}\log 12$

$pH = 10 + \frac{1}{3}[0.48 + 0.6]$ (1.08, 0.36)

$= 10.36$

A 500 mL saturated solution of $MgCO_3$ (M = 84) is reduced to 120 mL by evaporation. What mass of solid $MgCO_3$ is formed? [$K_{sp} = 4.0 \times 10^{-5}$]

(a) 0.013 g (b) 0.064 g
(c) 0.20 g (d) 0.27 g

$(s)^2 = 4 \times 10^{-5} = 4 \times 10 \times 10^{-6}$

$(s) = 2 \times 3.16 \times 10^{-2}$

$= 6.32 \times 10^{-3}$ M

$(s) \times [500 - 120] = (s) \times 380$ mmole

$= 6.32 \times 10^{-3} \times 0.380 \times 84$ gm

$= 0.2017$ gm